

CHAPTER 2

Causal Inference and Experimentation

Although the logic of experimentation is for the most part intuitive, researchers can run into trouble if they lack a firm grasp of the key assumptions that must be met in order for experiments to provide reliable assessments of cause and effect. This point applies in particular to field experimental researchers, who must frequently make real-time decisions about research design. Failure to understand core statistical principles and their practical implications may cause researchers to squander resources and experimental opportunities. It is wise, therefore, to invest time studying the formal statistical properties of experiments before launching a research project.

This chapter introduces a system of notation that will be used throughout the book. By depicting the outcomes that potentially manifest themselves depending on whether the treatment is administered to each unit, the notation clarifies a number of key concepts, such as the idea of a treatment effect. This notational system is then used to shed light on the conditions under which experiments provide persuasive evidence about cause and effect. The chapter culminates with a list of core assumptions and what they imply for experimental design. The advantage of working methodically from core principles is that a long list of design-related admonitions flows from a relatively compact set of ideas that can be stored in working memory.

2.1 Potential Outcomes

Suppose we seek to gauge the causal effect of a treatment. For concreteness, suppose we wish to study the budgetary consequences of having women, rather than men, head Indian village councils, which govern rural areas in West Bengal and Rajasthan.¹

¹ See Chatterjee and Duflo 2004.

What you will learn from this chapter:

1. The system of notation used to describe potential outcomes.
2. Definitions of core terms: average treatment effect, expectation, random assignment, and unbiasedness.
3. Assumptions that must be met in order for experiments to produce unbiased estimates of the average treatment effect.

Students of legislative politics have argued that women bring different policy priorities to the budgetary process in developing countries, emphasizing health issues such as providing clean drinking water. Leave aside for the time being the question of how this topic might be studied using randomly assigned treatments. For the moment, simply assume that each village either receives the treatment (a woman serves as village council head) or remains untreated (with its village council headed by a man). For each village, we also observe the share of the local council budget that is allocated to providing clean drinking water. To summarize, we observe the treatment (whether the village head is a woman or not) and the outcome (what share of the budget goes to a policy issue of special importance to women).

What we do not observe is how the budget in each village headed by a man would have been allocated if it had been headed by a woman, and vice versa. Although we do not observe these counterfactual outcomes, we can nevertheless imagine them. Taking this mental exercise one step further, we might imagine that each village has two *potential outcomes*: the budget it would enact if headed by a woman and the budget it would enact if headed by a man. The gender of the village head determines which potential budget we observe. The other budget remains imaginary or counterfactual.

Table 2.1 provides a stylized example of seven villages in order to introduce the notation that we will use throughout the book. The villages constitute the subjects in his experiment. Each subject is identified by a subscript i , which ranges from 1 to 7. The third village on the list, for example, would be designated as $i = 3$. The table imagines what would happen under two different scenarios. Let $Y_i(1)$ be the outcome if village i is exposed to the treatment (a woman as village head), and let $Y_i(0)$ be the outcome if this village is not exposed to the treatment. For example, Village 3 allocates 30% of its budget to water sanitation if headed by a woman but only 20% if headed by a man, so, $Y_3(1) = 30\%$, and $Y_3(0) = 20\%$. These are called potential outcomes because they describe what would happen if a treatment were or were not administered.

For purposes of this example, we assume that each village has just two potential outcomes, depending on whether it receives the treatment; villages are assumed to be unaffected by the treatments that other villages receive. In section 2.7, we spell out

TABLE 2.1

Illustration of potential outcomes for local budgets when village council heads are women or men. [Entries are shares of local budgets allocated to water sanitation.]

Village i	$Y_i(0)$ Budget share if village head is male	$Y_i(1)$ Budget share if village head is female	τ_i Treatment effect
Village 1	10	15	5
Village 2	15	15	0
Village 3	20	30	10
Village 4	20	15	-5
Village 5	10	20	10
Village 6	15	15	0
Village 7	15	30	15
Average	15	20	5

more precisely the assumptions that underlie the model of potential outcomes and discuss complications that arise when subjects are affected by the treatments that other subjects receive.

2.2 Average Treatment Effects

For each village, the causal effect of the treatment (τ_i) is defined as the difference between two potential outcomes:

$$\tau_i = Y_i(1) - Y_i(0). \quad (2.1)$$

In other words, the treatment effect for each village is the difference between two potential states of the world, one in which the village receives the treatment and another in which it does not. For Village 3, this causal effect is $30 - 20 = 10$.

The empirical challenge that researchers typically face when observing outcomes is that at any given time one can observe $Y_i(1)$ or $Y_i(0)$ but not both. (Bear in mind that the only reason we are able to see both potential outcomes for each village in Table 2.1 is that this is a hypothetical example!) Building on the notational system introduced above, we define Y_i as the observed outcome in each village and d_i as the observed treatment that is delivered in each village. In this case, Y_i is the observed share of the budget allocated to water sanitation, and d_i equals 1 when a woman is village head and 0 otherwise.

BOX 2.1**Potential Outcomes Notation**

In this system of notation, the subscript i refers to subjects 1 through N .

The variable d_i indicates whether the i th subject is treated: $d_i = 1$ means the i th subject receives the treatment, and $d_i = 0$ means the i th subject does not receive the treatment. It is assumed that d_i is observed for every subject.

$Y_i(1)$ is the potential outcome if the i th subject were treated. $Y_i(0)$ is the potential outcome if the i th subject were not treated. In general, potential outcomes may be written $Y_i(d)$, where d indexes the treatment. These potential outcomes are fixed attributes of each subject and represent the outcome that would be observed hypothetically if that subject were treated or untreated.

A schedule of potential outcomes refers to a comprehensive list of potential outcomes for all subjects. The rows of this schedule are indexed by i , and the columns are indexed by d . For example, in Table 2.1 the $Y_i(0)$ and $Y_i(1)$ potential outcomes for the fifth subject may be found in adjacent columns of the fifth row.

The connection between the observed outcome Y_i and the underlying potential outcomes is given by the equation $Y_i = d_i Y_i(1) + (1 - d_i) Y_i(0)$. This equation indicates that the $Y_i(1)$ are observed for subjects who are treated, and the $Y_i(0)$ are observed for subjects who are not treated. For any given subject, we observe either $Y_i(1)$ or $Y_i(0)$, never both.

It is sometimes useful to refer to potential outcomes for a subset of all subjects. Expressions of the form $Y_i(\cdot) | X = x$ denote potential outcomes when the condition $X = x$ holds. For example, $Y_i(0) | d_i = 1$ refers to the untreated potential outcome for a subject who actually receives the treatment.

Because we often want to know about the statistical properties of a hypothetical random assignment, we distinguish between d_i , the treatment that a given subject receives (a variable that one observes in an actual dataset), and D_i , the treatment that could be administered hypothetically. D_i is a random variable, and the i th subject might be treated in one hypothetical study and not in another. For example, $Y_i(1) | D_i = 1$ refers to the treated potential outcome for a subject who would be treated under some hypothetical allocation of treatments.

The budget that we observe in each village may be summarized using the following expression:

$$Y_i = d_i Y_i(1) + (1 - d_i) Y_i(0). \quad (2.2)$$

Because d_i is either 0 or 1, one of the terms on the right side of the equals sign will always be zero. We observe the potential outcome that results from treatment, $Y_i(1)$, if the treatment is administered ($d_i = 1$). If the treatment is not administered ($d_i = 0$), we observe the potential outcome that results when no treatment occurs, $Y_i(0)$.

The average treatment effect, or ATE, is defined as the sum of the τ_i divided by N , the number of subjects:

$$\text{ATE} = \frac{1}{N} \sum_{i=1}^N \tau_i. \quad (2.3)$$

An equivalent way to obtain the average treatment effect is to subtract the average value of $Y_i(0)$ from the average value of $Y_i(1)$:

$$\frac{1}{N} \sum_{i=1}^N Y_i(1) - \frac{1}{N} \sum_{i=1}^N Y_i(0) = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0)) = \frac{1}{N} \sum_{i=1}^N \tau_i. \quad (2.4)$$

The average treatment effect is an extremely important concept. Villages may have different τ_i , but the ATE indicates how outcomes would change on average if every village were to go from untreated (male village council head) to treated (female village council head).

From the rightmost column of Table 2.1, we can calculate the ATE for the seven villages. The average treatment effect in this example is 5 percentage points: if all villages were headed by men, they would on average spend 15% of their budgets on water sanitation, whereas if all villages were headed by women, this figure would rise to 20%.

BOX 2.2**Definition: Average Treatment Effect**

The average treatment effect (ATE) is the sum of the subject-level treatment effects, $Y_i(1) - Y_i(0)$, divided by the total number of subjects. An equivalent way to express the ATE is to say that it equals $\mu_{Y(1)} - \mu_{Y(0)}$, where $\mu_{Y(1)}$ is the average value of $Y_i(1)$ for all subjects and $\mu_{Y(0)}$ is the average value of $Y_i(0)$ for all subjects.

3 Random Sampling and Expectations

Suppose that instead of calculating the average potential outcome for all villages, we drew a random sample of villages and calculated the average among the villages we sampled. By *random sample*, we mean a selection procedure in which v villages are selected from the list of N villages, and every possible set of v villages is equally likely to be selected. For example, if we select one village at random from a list of seven villages, seven possible samples are equally likely. If we select three villages at random from a list of seven villages,

$$\frac{N!}{v!(N-v)!} = \frac{7!}{3!4!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1)(4 \times 3 \times 2 \times 1)} = 35 \quad (2.5)$$

possible samples are equally likely. If potential outcomes vary from one village to the next, the average potential outcome in the villages we sample will vary, depending on which of the possible samples we happen to select. The sample average may be characterized as a *random variable*, a quantity that varies from sample to sample.

The term *expected value* refers to the average outcome of a random variable. (See Box 2.3.) In our example, the random variable is the number we obtain when we sample villages at random and calculate their average outcome. Recall from introductory statistics that under random sampling, the expected value of a sample average is equal to the average of the population from which the sample is drawn.² This principle may be illustrated using the population of villages depicted in Table 2.1. Recall that the average value of $Y_i(0)$ among all villages in Table 2.1 is 15. Suppose we sample two villages at random from the list of seven villages and calculate the average value of $Y_i(0)$ for the two selected villages. There are

$$\frac{N!}{v!(N-v)!} = \frac{7!}{2!5!} = 21 \quad (2.6)$$

possible ways of sampling two villages at random from a list of seven, and each sample is equally likely to be drawn. Any given sample of two villages might contain an average value of $Y_i(0)$ that is higher or lower than the true average of 15, but the expected value refers to what we would obtain on average if we were to examine all 21 possible samples, for each one calculating the average value of $Y_i(0)$:

$$\{10, 12.5, 12.5, 12.5, 12.5, 12.5, 15, 15, 15, 15, 15, 15, 15, 17.5, 17.5, 17.5, 17.5, 17.5, 17.5, 20\}. \quad (2.7)$$

² The easiest way to see the intuition behind this principle is to consider the case in which we randomly sample just one village. Each village is equally likely to be sampled. The average over all seven possible samples is identical to the average for the entire population of seven villages. This logic generalizes to samples where $v > 1$ because each village appears in exactly $v/7$ of all possible samples.

BOX 2.3

The expectation of a discrete random variable X is defined as

$$E[X] = \sum x \Pr[X = x],$$

where $\Pr[X = x]$ denotes the probability that X takes on the value x , and where the summation is taken over all possible values of x .

For example, what is the expected value of a randomly selected value of τ_i from Table 2.1?

$$\begin{aligned} E[\tau_i] &= \sum \tau \Pr[\tau_i = \tau] \\ &= (-5)\left(\frac{1}{7}\right) + (0)\left(\frac{2}{7}\right) + (5)\left(\frac{1}{7}\right) + (10)\left(\frac{2}{7}\right) + (15)\left(\frac{1}{7}\right) = 5. \end{aligned}$$

Properties of Expectations

The expectation of the constant α is itself: $E[\alpha] = \alpha$.

For a random variable X and constants α and β , $E[\alpha + \beta X] = \alpha + \beta E[X]$.

The expectation of a sum of two random variables, X and Y , is the sum of their expectations: $E[X + Y] = E[X] + E[Y]$.

The expectation of the product of two random variables, X and Y , is the product of their expectations plus the covariance between them: $E[XY] = E[X]E[Y] + E[(X - E[X])(Y - E[Y])]$.

The average of these 21 numbers is 15. In other words, the expected value of the average $Y_i(0)$ obtained from a random sample of two villages is 15.

The concept of expectations plays an important role in the discussion that follows. Because we will refer to expectations so often, a bit more notation is helpful. The notation $E[X]$ refers to the expectation of a random variable X . (See Box 2.3.) The expression “the expected value of $Y_i(0)$ when one subject is sampled at random” will be written compactly as $E[Y_i(0)]$. When a term like $Y_i(0)$ appears in conjunction with an expectations operator, it should be read not as the value of $Y_i(0)$ for subject i but instead as a random variable that is equal to the value of $Y_i(0)$ for a randomly selected subject. When the expression $E[Y_i(0)]$ is applied to values in Table 2.1, the random variable is the random selection of a $Y_i(0)$ from the list of all $Y_i(0)$; since there are seven possible random selections, the average of which is 15, it follows that $E[Y_i(0)] = 15$.

Sometimes attention is focused on the expected value of a random variable within a subgroup. *Conditional expectations* refer to subgroup averages. In terms of notation, the logical conditions following the $|$ symbol indicate the criteria that define the subgroup. For example, the expression “the expectation of $Y_i(1)$ when the village is selected at random from those villages that were treated” is written $E[Y_i(1) | d_i = 1]$. The idea of a conditional expectation is straightforward when working with quantities that are in principle observable. More mind-bending are expressions like $E[Y_i(1) | d_i = 0]$, which denotes “the expectation of $Y_i(1)$ when one village is selected at random from those villages that were not treated.” In the course of conducting research, we will never actually see $Y_i(1)$ for an untreated village, nor will we see $Y_i(0)$ for a treated village. These potential outcomes can be imagined but not observed.

One special type of conditional expectation arises when the subgroup is defined by the outcome of a random process. In that case, the conditional expectation may vary depending on which subjects happened to meet the condition in any particular realization of the random process. For example, suppose that a random process, such as a coin flip, determines which subjects are treated. For a given treatment assignment d_i , we could calculate $E[Y_i(1) | d_i = 0]$, but this expectation might have been different had the coin flips come out differently. Suppose we want to know the expected conditional expectation, or how the conditional expectation would come out, on average, across all possible ways that d_i could have been allocated. Let D_i be a random variable that indicates whether each subject would be treated in a hypothetical experiment. The conditional expectation $E[Y_i(1) | D_i = 0]$ is calculated by considering all possible realizations of D_i (all the possible ways that N coins could have been flipped) in order to form the joint probability distribution function for $Y_i(1)$ and D_i . As long as we know the joint probability of observing each paired set of values $\{Y(1), D\}$, we can calculate the conditional expectation using the formula in Box 2.4.³

With this basic system of notation in place, we may now describe the connection between expected potential outcomes and the average treatment effect (ATE):

$$\begin{aligned} E[Y_i(1) - Y_i(0)] &= E[Y_i(1)] - E[Y_i(0)] \\ &= \frac{1}{N} \sum_{i=1}^N Y_i(1) - \frac{1}{N} \sum_{i=1}^N Y_i(0) \\ &= \frac{1}{N} \sum_{i=1}^N [Y_i(1) - Y_i(0)] = \text{ATE}. \end{aligned} \quad (2.8)$$

³ The notation $E[Y_i(1) | D_i = 0]$ may be regarded as shorthand for $E[E[Y_i(1) | d_i = 0, d]]$, where d refers to a vector of treatment assignments and d_i refers to its i th element. Given d , we may calculate the probability distribution function for all $\{Y(1), d\}$ pairs and the expectation given this set of assignments. Then we may

BOX 2.4

Definition: Conditional Expectation

For discrete random variables Y and X , the conditional expectation of Y given that X takes on the value x is

$$E[Y | X = x] = \sum_y y \Pr[Y = y | X = x] = \sum_y \frac{\Pr[Y = y, X = x]}{\Pr[X = x]},$$

where $\Pr[Y = y, X = x]$ denotes the joint probability of $Y = y$ and $X = x$, and where the summation is taken over all possible values of y .

For example, in Table 2.1 what is the conditional expectation of a randomly selected value of τ_i for villages where $Y_i(0) > 10$? This question requires us to describe the joint probability distribution function for the variables τ_i and $Y_i(0)$ so that we can calculate $\Pr[\tau_i = \tau, Y_i(0) > 10]$. Table 2.1 indicates that the $\{\tau, Y(0)\}$ pair $\{0, 15\}$ occurs with probability $2/7$, while the other pairs $\{5, 10\}$, $\{10, 20\}$, $\{-5, 20\}$, $\{10, 10\}$, and $\{15, 15\}$ each occur with probability $1/7$. The marginal distribution of $Y_i(0)$ reveals that 5 of the 7 $Y_i(0)$ are greater than 10, so $\Pr[Y_i(0) > 10] = 5/7$.

$$\begin{aligned} E[\tau_i | Y_i(0) > 10] &= \sum \tau \frac{\Pr[\tau_i = \tau, Y_i(0) > 10]}{\Pr[Y_i(0) > 10]} \\ &= (-5)\frac{1}{7} + (0)\frac{2}{7} + (5)\frac{0}{7} + (10)\frac{1}{7} + (15)\frac{1}{7} = 4. \end{aligned}$$

In order to illustrate the idea of a conditional expectation when conditioning on the outcome of a random process, suppose we randomly assign one of the observations in Table 2.1 to treatment ($D_i = 1$) and the remaining six observations to control ($D_i = 0$). If each of the seven possible assignments occurs with probability $1/7$, what is the expected value of a randomly selected τ_i given that $D_i = 1$? Again, we start with the joint probability density function for τ_i and D_i and consider all possible pairings of these two variables' values. The $\{\tau, D\}$ pairings $\{-5, 1\}$, $\{5, 1\}$, and $\{15, 1\}$ occur with probability $1/49$, while the pairings $\{0, 1\}$ and $\{10, 1\}$ occur with probability $2/49$; the remaining $\{\tau, D\}$ pairings are instances in which τ is paired with 0. The marginal distribution $\Pr[D_i = 1] = 3(1/49) + 2(2/49) = 1/7$.

$$\begin{aligned} E[\tau_i | D_i = 1] &= \sum \tau \frac{\Pr[\tau_i = \tau, D_i = 1]}{\Pr[D_i = 1]} \\ &= (-5)\frac{1}{49} + (0)\frac{2}{49} + (5)\frac{1}{49} + (10)\frac{2}{49} + (15)\frac{1}{49} = 5. \end{aligned}$$

he first line of equation (2.8) expresses the fact that when a village is selected at random from the list of villages, its expected treatment effect is equal to the difference between the expected value of a randomly selected treated potential outcome and the expected value of a randomly selected untreated potential outcome. The second equality in equation (2.8) indicates that the expected value of a randomly selected $Y_i(1)$ equals the average of all $Y_i(1)$ values, and that the expected value of a randomly selected $Y_i(0)$ equals the average of all $Y_i(0)$ values. The third equality reflects the fact that the difference between the two averages in the second line of equation (2.8) can be expressed as the average difference in potential outcomes. The final equality notes that the average difference in potential outcomes is the definition of the average treatment effect. In sum, the difference in expectations equals the difference in average potential outcomes for the entire list of villages, or the ATE.⁴

This relationship is apparent from the schedule of potential outcomes in Table 2.1. The column of numbers representing the treatment effect (τ_i) is, on average, 5. If we were to select villages at random from this list, we would expect their average treatment effect to be 5. We get the same result if we subtract the expected value of a randomly selected $Y_i(0)$ from the expected value of a randomly selected $Y_i(1)$.

2.4 Random Assignment and Unbiased Inference

The challenge of estimating the average treatment effect is that at a given point in time each village is either treated or not: either $Y_i(1)$ or $Y_i(0)$ is observed, but not both. To illustrate the problem, Table 2.2 shows what outcomes would be observed if Village 1 and Village 7 were treated, while the remaining villages were not. We observe $Y_i(1)$ for Villages 1 and 7 but not $Y_i(0)$. For Villages 2, 3, 4, 5, and 6, we observe $Y_i(0)$ but not $Y_i(1)$. The unobserved or “missing” values in Table 2.2 are indicated with a “?”.

The notation used here is just one way to explicate the link between expectations and the ATE. Samii and Ronow (2012) suggest an alternative formalization. Their model envisions a finite population U consisting of units j in $1, 2, \dots, N$, each of which has an associated triple $(y_j(1), y_j(0), D_j)$ such that $y_j(1)$ and $y_j(0)$ are fixed potential outcomes and D_j is a random variable indicating the treatment status of unit j . Reassign random index ordering i in $1, 2, \dots, N$. Then, for an arbitrary unit i , there exists an associated triple of random variables $(Y_i(1), Y_i(0), D_i)$ such that the random variable $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. It follows that for equation (2.8):

$$E[Y_i(1)] - E[Y_i(0)] = \frac{1}{N} \sum_{j=1}^N y_j(1) - \frac{1}{N} \sum_{j=1}^N y_j(0) = \text{ATE}.$$

Statistical operators such as expectations or independence refer to random variables associated with an arbitrary index i . Looking ahead to later chapters, one might expand this system to include other unit-level attributes, such as covariates or missingness, by attaching them to the triple indexed by j before reassigning the ordering.

TABLE 2.2

Illustration of observed outcomes for local budgets when two village councils are headed by women.

Village i	$Y_i(0)$ Budget share if village head is male	$Y_i(1)$ Budget share if village head is female	τ_i Treatment effect
Village 1	?	15	?
Village 2	15	?	?
Village 3	20	?	?
Village 4	20	?	?
Village 5	10	?	?
Village 6	15	?	?
Village 7	?	30	?
Estimated average based on observed data	16	22.5	6.5

Note: The observed outcomes in this table are based on the potential outcomes listed in Table 2.1.

Random assignment addresses the “missing data” problem by creating two groups of observations that are, in expectation, identical prior to application of the treatment. When treatments are allocated randomly, the treatment group is a random sample of all villages, and therefore the expected potential outcomes among villages in the treatment group are identical to the average potential outcomes among all villages. The same is true for villages in the control group. The control group’s expected potential outcomes are also identical to the average potential outcomes among all villages. Therefore, in expectation, the treatment group’s potential outcomes are the same as the control group’s. Although any given random allocation of villages to treatment and control groups may produce groups of villages that have different average potential outcomes, this procedure is fair in the sense that it does not tend to give one group a higher set of potential outcomes than the other.

As Chattopadhyay and Duflo point out, random assignment is in fact used in rural India to assign women to head one-third of the local village councils.⁵ Ordinarily, men would head the village councils, but Indian law mandates that selected

5 Chattopadhyay and Duflo 2004. A lottery is used to assign council positions to women in Rajasthan. In West Bengal, a near-random assignment procedure is used whereby villagers are assigned according to their serial numbers.

villages install a female representative as head of the council. For purposes of illustration, suppose that our collection of seven villages were subject to this law, and that 10 villages will be randomly assigned female council heads. Consider the statistical implications of this arrangement. This random assignment procedure implies that every village has the same probability of receiving the treatment; assignment bears no systematic relationship to villages' observed or unobserved attributes.

Let's take a closer look at the formal implications of this form of random assignment. When villages are assigned such that every village has the same probability of receiving the treatment, the villages that are randomly chosen for treatment are a random subset of the entire set of villages. Therefore, the expected $Y_i(1)$ potential outcome among treated villages is the same as the expected $Y_i(1)$ potential outcome for the entire set of villages:

$$E[Y_i(1) | D_i = 1] = E[Y_i(1)]. \quad (2.9)$$

BOX 2.5

Two Commonly Used Forms of Random Assignment

Random assignment refers to a procedure that allocates treatments with known probabilities that are greater than zero and less than one.

The most basic forms of random assignment allocate treatments such that every subject has the same probability of being treated. Let N be the number of subjects, and let m be the number of subjects who are assigned to the treatment group. Assume that N and m are integers such that $0 < m < N$. Simple random assignment refers to a procedure whereby each subject is allocated to the treatment group with probability m/N . Complete random assignment refers to a procedure that allocates exactly m units to treatment.

Under simple or complete random assignment, the probability of being assigned to the treatment group is identical for all subjects; therefore treatment status is statistically independent of the subjects' potential outcomes and their background attributes (X):

$$Y_i(0), Y_i(1), X \perp\!\!\!\perp D_i$$

where the symbol $\perp\!\!\!\perp$ means "is independent of." For example, if a die roll is used to assign subjects to treatment with probability $1/6$, knowing whether a subject is treated provides no information about the subject's potential outcomes or background attributes. Therefore, the expected value of $Y_i(0)$, $Y_i(1)$, and X_i is the same in treatment and control groups.

When we randomly select villages into the treatment group, the villages we leave behind for the control group are also a random sample of all villages. The expected $Y_i(1)$ in the control group ($D_i = 0$) is therefore equal to the expected $Y_i(1)$ for the entire set of villages:

$$E[Y_i(1) | D_i = 0] = E[Y_i(1)]. \quad (2.10)$$

Putting equations (2.9) and (2.10) together, we see that under random assignment the treatment and control groups have the same expected potential outcome:

$$E[Y_i(1) | D_i = 1] = E[Y_i(1) | D_i = 0]. \quad (2.11)$$

Equation (2.11) also underscores the distinction between realized and unrealized potential outcomes. On the left side of the equation is the expected treated potential outcome among villages that receive the treatment. The treatment causes this potential outcome to become observable. On the right side of the equation is the expected treated potential outcome among villages that do not receive the treatment. Here, the lack of treatment means that the treated potential outcome remains unobserved for these subjects.

The same logic applies to the control group. Villages that do not receive the treatment ($D_i = 0$) have the same expected untreated potential outcome $Y_i(0)$ that the treatment group ($D_i = 1$) would have if it were untreated:

$$E[Y_i(0) | D_i = 0] = E[Y_i(0) | D_i = 1] = E[Y_i(0)]. \quad (2.12)$$

Equations (2.11) and (2.12) follow from random assignment: D_i conveys no information whatsoever about the potential values of $Y_i(1)$ or $Y_i(0)$. The randomly assigned values of D_i determine which value of Y_i we actually *observe*, but they are nevertheless statistically independent of the *potential* outcomes $Y_i(1)$ and $Y_i(0)$. (See Box 2.5 for discussion of the term *independence*.)

When treatments are assigned randomly, we may rearrange equations (2.8), (2.11), and (2.12) in order to express the average treatment effect as

$$ATE = E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0]. \quad (2.13)$$

This equation suggests an empirical strategy for estimating the average treatment effect. The terms $E[Y_i(1) | D_i = 1]$ and $E[Y_i(0) | D_i = 0]$ may be estimated using experimental data. We do not observe the $Y_i(1)$ potential outcomes for all observations, but we do observe them for the random sample of observations that receive the treatment. Similarly, we do not observe the $Y_i(0)$ potential outcomes for all observations, but we do observe them for the random sample of observations in the control group. If we want to estimate the average treatment effect, equation (2.13) suggests that we should take the difference between two sample means: the average

come in the treatment group minus the average outcome in the control group, as that enable researchers to use observable quantities (e.g., sample averages) to infer parameters of interest (e.g., average treatment effects) are termed *identification strategies*.

Statistical procedures used to make guesses about parameters such as the average treatment effect are called *estimators*. In this example, the estimator is very simple, just a difference between two sample averages. Before applying an estimator to actual data, a researcher should reflect on its statistical properties. One especially important property is *unbiasedness*. An estimator is unbiased if it generates the right answer, on average. In other words, if the experiment were replicated an infinite number of times under identical conditions, the average *estimate* would equal the true parameter. Some guesses may be too high and others too low, but the average guess will be correct. In practice, we will not be able to perform an infinite number of experiments. In fact, we might just perform one experiment and leave it at that. Nevertheless, in theory we can analyze the properties of our estimation procedure to see whether, on average, it recovers the right answer. (In the next chapter, we consider another property of estimators: how precisely they estimate the parameter of interest.)

In sum, when treatments are administered using a procedure that gives every subject the same probability of being treated, potential outcomes are independent of the treatments that subjects receive. This property suggests an identification strategy for estimating average treatment effects using experimental data.

The remaining task is to demonstrate that the proposed estimator—the difference between the average outcome in the treatment group and the average outcome in the control group—is an unbiased estimator of the ATE when all subjects have the same probability of being treated. The proof is straightforward. Because the units assigned to the control group are a random sample of all units, the average of the control group outcomes is an unbiased estimator of the average value of $Y_i(0)$

BOX 2.6

Definition: Estimator and Estimate

An estimator is a procedure or formula for generating guesses about parameters such as the average treatment effect. The guess that an estimator generates based on a particular experiment is called an estimate. Estimates are denoted using a “hat” notation. The estimate of the parameter θ is written $\hat{\theta}$.

among all units. The same goes for the treatment group: the average outcome among units that receive the treatment is an unbiased estimator of the average value of $Y_i(1)$ among all units. Formally, if we randomly shuffle the villages and place the first m subjects in the treatment group and the remaining $N - m$ subjects in the control group, we can analyze the expected, or average, outcome over all possible random assignments:

$$\begin{aligned} E \left[\frac{\sum_{i=1}^m Y_i}{m} - \frac{\sum_{i=m+1}^N Y_i}{N-m} \right] &= E \left[\frac{\sum_{i=1}^m Y_i}{m} \right] - E \left[\frac{\sum_{i=m+1}^N Y_i}{N-m} \right] \\ &= \frac{E[Y_1] + E[Y_2] + \cdots + E[Y_m]}{m} - \frac{E[Y_{m+1}] + E[Y_{m+2}] + \cdots + E[Y_N]}{N-m} \\ &= E[Y_i(1) | D_i = 1] - E[Y_i(0) | D_i = 0] \\ &= E[Y_i(1)] - E[Y_i(0)] = E[\tau_i] = \text{ATE}. \end{aligned} \quad (2.14)$$

Equation (2.14) conveys a simple but extremely useful idea. When units are randomly assigned, a comparison of average outcomes in treatment and control groups (the so-called *difference-in-means estimator*) is an unbiased estimator of the average treatment effect.

BOX 2.7

Definition: Unbiased Estimator

An estimator is unbiased if the expected value of the estimates it produces is equal to the true parameter of interest. Call θ the parameter we seek to estimate, such as the ATE. Let $\hat{\theta}$ represent an estimator, or procedure for generating estimates. For example, $\hat{\theta}$ may represent the difference in average outcomes between treatment and control groups. The expected value of this estimator is the average estimate we would obtain if we apply this estimator to all possible realizations of a given experiment or observational study. We say that $\hat{\theta}$ is unbiased if $E(\hat{\theta}) = \theta$; in words, the estimator $\hat{\theta}$ is unbiased if the expected value of this estimator is θ , the parameter of interest. Although unbiasedness is a property of estimators and not estimates, we refer to the estimates generated by an unbiased estimator as “unbiased estimates.”

5 The Mechanics of Random Assignment

The result in equation (2.14) hinges on random assignment, and so it is important to be clear about what constitutes random assignment. *Simple random assignment* is a term of art, referring to a procedure—a die roll or coin toss—that gives each subject an identical probability of being assigned to the treatment group. The practical drawback of simple random assignment is that when N is small, random chance can create a treatment group that is larger or smaller than what the researcher intended. For example, you could flip a coin to assign each of 10 subjects to the treatment condition, but there is only a 24.6% chance of ending up with exactly 5 subjects in treatment and 5 in control. A useful special case of simple random assignment is *complete random assignment*, where exactly m of N units are assigned to the treatment group with equal probability.⁶

The procedure used to conduct complete random assignment can take any of three equivalent forms. Suppose one has N subjects and seeks to assign treatments to m of them. The first method is to select one subject at random, then select another random from the remaining units, and so forth until you have selected m subjects to the treatment group. A second method is to enumerate all of the possible ways that m subjects may be selected from a list of N subjects, and randomly select one of the possible allocation schemes. A third method is to randomly permute the order of the N subjects and label the first m subjects as the treatment group.⁷

Beware of the fact that *random* is a word that is used loosely in common parlance to refer to procedures that are arbitrary, haphazard, or unplanned. The problem is that arbitrary, haphazard, or unplanned treatments may follow systematic patterns that go unnoticed. Procedures such as alternation are risky because there may be systematic reasons why certain types of subjects might alternate in a sequence, and indeed, some early medical experiments ran into exactly this problem.⁸ We use the term *random* in a more exacting sense. The physical or electronic procedure by which randomization is conducted ensures that assignment to the treatment group is statistically independent of all observed or unobserved variables.

⁶ In Chapters 3 and 4, we discuss other frequently used methods of random assignment: clustered random assignment, where groups of subjects are randomly assigned to treatment and control, and block random assignment (also called stratified random assignment), where individuals are first divided into blocks, and then random assignment is performed within each block. Box 2.5 notes that a defining feature of complete (as opposed to clustered or blocked) random assignment is that all possible assignments of m subjects to a treatment group of size m are equally likely.

⁷ Cox and Reid 2000, p. 20. The term *complete randomization* is a bit awkward, as the word *complete* does not convey the requirement that exactly m units are allocated to treatment, but this terminology has become standard (see Rosenbaum 2002, pp. 25–26).

⁸ Torgerson, Gatzsche and Ghind 1998.

In practical terms, random assignment is best done using statistical software. Here is an easy procedure for implementing complete random assignment. First, determine N , the number of subjects in your experiment, and m , the number of subjects who will be allocated to the treatment group. Second, set a random number “seed” using a statistics package, so that your random numbers may be reproduced by anyone who cares to replicate your work. Third, generate a random number for each subject. Fourth, sort the subjects by their random numbers in ascending order. Finally, classify the first m observations as the treatment group. Example programs using R may be found at <http://isps.research.yale.edu/FEDAI>.

Generating random numbers is just the first step in implementing random assignment. After the numbers are generated, one must take pains to preserve the integrity of the assignment process. A deficiency of alternation and many other arbitrary procedures is that they allow those administering the allocation to foresee who will be assigned to which experimental group. If a receptionist seeks to get the sickest patients into the experimental treatment group and knows that the pattern of assignments alternates, he can reorder the patients in such a way as to shuttle the sickest subjects into the treatment group.⁹ The same concern arises even when a random sequence of numbers is used to assign incoming patients: random allocation may be undone if the receptionist knows the order of assignments ahead of time, because that enables him to position patients so that they will be assigned to a certain experimental group. In order to guard against potential threats to the integrity of random assignment, researchers should build extra procedural safeguards into their experimental designs, such as blinding those administering the experiment to the subjects’ assigned experimental groups.

2.6 The Threat of Selection Bias When Random Assignment Is Not Used

Without random assignment, the identification strategy derived from equation (2.14) unravels. The treatment and control groups are no longer random subsets of all units in the sample. Instead, we confront what is known as a *selection problem*: receiving treatment may be systematically related to potential outcomes. For example, absent random assignment, villages determine whether their councils are headed by women. The villages that end up with female council heads may not be a random subset of all villages.

⁹ For examples of experiments in which random assignment was subverted, see Torgerson and Torgerson 2008.

To see how nonrandom selection jeopardizes the identification strategy of comparing average outcomes in the treatment and control groups, rewrite the expected difference in outcomes from equation (2.13) by subtracting and adding $E[Y_i(0)|D_i = 1]$:

$$\begin{aligned} & \underbrace{E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{Expected difference between treated and untreated outcomes}} \\ &= \underbrace{E[Y_i(1) - Y_i(0)|D_i = 1]}_{\text{ATE among the treated}} + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{Selection bias}}. \quad (2.15) \end{aligned}$$

Under random assignment, the selection bias term is zero, and the ATE among the (randomly) treated villages is the same as the ATE among all villages. In the absence of random assignment, equation (2.15) warns that the apparent treatment effect is a mixture of selection bias and the ATE for a subset of villages.

In order to appreciate the implications of equation (2.15), consider the following scenario. Suppose that instead of randomly selecting villages to receive the treatment, our procedure were to let villages decide whether to take the treatment. Refer back to Table 2.1 and imagine that, if left to their own devices, Village 5 and Village 7 always elect a woman due to villagers' pent-up demand for water sanitation, while the remaining villages always elect a man.¹⁰ Self-selection in this case leads to an exaggerated estimate of the ATE because receiving the treatment is associated with lower-than-average values of $Y_i(0)$ and higher-than-average values of $Y_i(1)$. The average outcome in the treatment group is 25, and the average outcome in the control group is 16. The estimated ATE is therefore 9, whereas the actual ATE is 5. Referring to equation (2.15) we see that in this case the ATE among the treated is not equal to the ATE for the entire subject pool, nor is the selection bias term equal to zero. The broader point is that it is risky to compare villages that choose to receive the treatment with villages that choose not to. In this example, self-selection is related to potential outcomes; as a result, the comparison of treated and untreated villages recovers neither the ATE for the sample as a whole nor the ATE among those villages that receive treatment.

The beauty of experimentation is that the randomization procedure generates a schedule of treatment and control assignments that are statistically independent of

potential outcomes. In other words, the assumptions underlying equations (2.9) to (2.13) are justified by reference to the *procedure* of random assignment, not substantive arguments about the comparability of potential outcomes in the treatment and control groups.

The preceding discussion should not be taken to imply that experimentation invokes no substantive assumptions. The unbiasedness of the difference-in-means estimator hinges not only on random assignment but also on two assumptions about potential outcomes, the plausibility of which will vary depending on the application. The next section spells out these important assumptions.

2.7 Two Core Assumptions about Potential Outcomes

To this point, our characterization of potential outcomes has glossed over two important details. In order to ease readers into the framework of potential outcomes, we simply stipulated that each subject has two potential outcomes, $Y_i(1)$ if treated and $Y_i(0)$ if not treated. To be more precise, each potential outcome depends *solely* on whether the subject *itself* receives the treatment. When writing potential outcomes in this way, we are assuming that potential outcomes respond only to the treatment and not some other feature of the experiment, such as the way the experimenter assigns treatments or measures outcomes. Furthermore, potential outcomes are defined over the set of treatments that the subject *itself* receives, not the treatments assigned to other subjects. In technical parlance, the “solely” assumption is termed *excludability* and the “itself” assumption is termed *non-interference*.

2.7.1 Excludability

When we define two, and only two, potential outcomes based on whether the treatment is administered, we implicitly assume that the only relevant causal agent is receipt of the treatment. Because the point of an experiment is to isolate the causal effect of the treatment, our schedule of potential outcomes excludes from consideration factors other than the treatment. When conducting an experiment, therefore, we must define the treatment and distinguish it from other factors with which it may be correlated. Specifically, we must distinguish between d_i , the treatment, and z_i , a variable that indicates which observations have been allocated to treatment or control. We seek to estimate the effect of d_i , and we assume that the treatment assignment z_i has no effect on outcomes except insofar as it affects the value of d_i .

The term *exclusion restriction* or *excludability* refers to the assumption that z_i can be omitted from the schedule of potential outcomes for $Y_i(1)$ and $Y_i(0)$. Formally, this

¹⁰ When taking expectations over hypothetical replications of an experiment, we consider all possible random assignments. In our example of non-random allocation, however, nature makes the assignment. When taking expectations, we must therefore consider the average of all possible natural assignments. Rather than make up an assortment of possible assignments and stipulate the probability that each scenario occurs, we have kept the example as simple as possible and assumed that the villages “always” elect the same type of candidate. In effect, we are taking expectations over just one possible assignment that occurs with probability 1.

assumption may be written as follows. Let $Y_i(z, d)$ be the potential outcome when $z = z$ and $d_i = d$, for $z \in (0, 1)$ and for $d \in (0, 1)$. For example, if $z_i = 1$ and $d_i = 1$, the subject is assigned to the treatment group and receives the treatment. We can also envision other combinations. For example, if $z_i = 1$ and $d_i = 0$, the subject is assigned to the treatment group but for some reason does not receive the treatment. The exclusion restriction assumption is that $Y_i(1, d) = Y_i(0, d)$. In other words, potential outcomes respond only to the input from d_i ; the value of z_i is irrelevant. Unfortunately, this assumption cannot be verified empirically because we never observe both $Y_i(1, d)$ and $Y_i(0, d)$ for the same subject.

The exclusion restriction breaks down when random assignment sets in motion causes of Y_i other than the treatment d_i . Suppose the treatment in our running example were defined as whether or not a woman council head presides over deliberations about village priorities. Our ability to estimate the effect of this treatment would be jeopardized if nongovernmental aid organizations, sensing that newly elected women will prioritize clean water, were to redirect their efforts to promote water sanitation in male-led villages. If outside aid flows to male-led villages, obviating the need for male village council leaders to allocate their budgets to water sanitation, the apparent difference between water sanitation budgets in councils led by women and councils led by men will exaggerate the true effect of the treatment, as defined above.¹¹ Even if it were the case that women council leaders have no effect on their own villages' budgets, the behavior of the NGOs could generate different average budgets in male- and female-led villages.

Asymmetries in measurement represent another threat to the excludability assumption. Suppose, for example, that in our study of Indian villages, we were to dispatch one group of research assistants to measure budgets in the treatment group and a different group of assistants to measure budgets in the control group. Each group of assistants may apply a different standard when determining what expenditures are to be classified as contributing to water sanitation. Suppose the research assistants in the treatment group were to use a more generous accounting standard—they tend to exaggerate the amount of money that the village allocates to water sanitation. When we compare average budgets in the treatment and control groups, the estimated treatment effect will be a combination of the true effect of female village heads on budgets and accounting procedures that exaggerate the amount of money spent on water sanitation in those villages. Presumably, when we envisioned the experiment and what we might learn from it, we sought to estimate only the first of these two effects. We wanted to know the effect of female leadership on budgets using a consistent standard of accounting.

11 Whether an excludability violation occurs depends on how a treatment effect is defined. If one were to define the effect of electing a woman to include the compensatory behavior of NGOs, this assumption

To illustrate the consequences of measurement asymmetry, we may write out a simple model in which outcomes are measured with error. Under this scenario, the usual schedule of potential outcomes expands to reflect the fact that outcomes are influenced not only by d_i , but also by z_i , which determines which set of research assistants measure the outcome. Suppose that among untreated units we observe $Y_i(0)^* = Y_i(0) + e_{i0}$, where e_{i0} is the error that is made when measuring the potential outcome if an observation is assigned to the control group. For treated units, let $Y_i(1)^* = Y_i(1) + e_{i1}$. What happens if we compare average outcomes among treated and untreated units? The expected value of the difference-in-means estimator from equation (2.14) is

$$E\left[\frac{\sum_1^m Y_i}{m} - \frac{\sum_{m+1}^N Y_i}{N-m}\right] = E[Y_i(1)^* | D_i = 1] - E[Y_i(0)^* | D_i = 0] \\ = E[Y_i(1) | D_i = 1] + E[e_{i1} | D_i = 1] - E[Y_i(0) | D_i = 0] - E[e_{i0} | D_i = 0]. \quad (2.16)$$

Comparing equation (2.16) to equation (2.14) reveals that the difference-in-means estimator is biased when the measurement errors in the treated and untreated groups have different expected values:

$$E[e_{i1} | D_i = 1] \neq E[e_{i0} | D_i = 0]. \quad (2.17)$$

In this book, when we speak of a "breakdown in symmetry," we have in mind procedures that may distort the expected difference between treatment and control outcomes.

What kinds of experimental procedures bolster the plausibility of the excludability assumption? The broad answer is anything that helps ensure uniform handling of treatment and control groups. One type of procedure is double-blindness—neither the subjects nor the researchers charged with measuring outcomes are aware of which treatments the subjects receive, so that they cannot consciously or unconsciously distort the results. Another procedure is parallelism in the administration of an experiment: the same questionnaires and survey interviewers should be used to assess outcomes in both treatment and control groups, and both groups' outcomes should be gathered at approximately the same time and under similar conditions. If outcomes for the control group are gathered in October, but outcomes in the treatment group are gathered in November, symmetry may be jeopardized.

The exclusion restriction cannot be evaluated unless the researcher has stated precisely what sort of treatment effect the experiment is intended to measure and designed the experiment accordingly. Depending on the researcher's objective, the control group may receive a special type of treatment so that the treatment vs. control comparison isolates a particular aspect of the treatment. A classic example of a research design that attempts to isolate a specific cause is a pharmaceutical trial in

ich an experimental pill is administered to the treatment group while an identical sugar pill is administered to the control group. The aim of administering a pill to both groups is to isolate the pharmacological effects of the ingredients, holding constant the effect of merely taking some sort of pill. In the village council example, a researcher may wish to distinguish the effects of female leadership of local councils from the effects of merely appointing non-incumbents to the headship. In principle, one could compare districts with randomly assigned women heads to districts with randomly assigned term limits, a policy that has the effect of bringing non-incumbents into leadership roles. This approach to isolating causal mechanisms is revisited again in Chapter 10, where we discuss designs that attempt to differentiate the active ingredients in a multifaceted treatment.

Protecting the theoretical integrity of the treatment vs. control comparison is of paramount importance in experimental design. In the case of the village budget study, the aim is to estimate the budgetary consequences of having a randomly allocated female village head, not the consequences of using a different measurement standard to evaluate outcomes in treatment and control villages. The same argument goes for other aspects of research activity that might be correlated with treatment assignment. For example, if the aim is to measure the effect of female leadership on budgets per village, bias may be introduced if one sends a delegation of researchers to monitor village council deliberations in women-headed villages only. Now the observed treatment effect is a combination of the effect of female leadership and the effect of research resources. Whether one regards the presence of the research delegation as a distortion of measurement or an unintended pathway by which assignment to treatment affects the outcome, the formal structure of the problem remains the same. The expected outcome of the experiment no longer reveals the causal effect we set out to estimate.

The symmetry requirement does not rule out cross-cutting treatments. For example, one could imagine a version of India's reservation policy that randomly assigned some village council seats to women, others to people from lower castes, and still others to women from lower castes. When we discuss factorial designs in Chapter 9, we will stress what can be learned from deploying several treatments in combination with one another. The point of these more complex designs is to learn about combinations of treatments while still preserving symmetry: randomly assigning treatments both alone and in combination with one another allows the researcher to distinguish empirically between having a female village head and having a female village head who is also from a lower caste.

Finally, let's revisit the case in which other actors intervene in response to your treatment assignments. For example, suppose that in anticipation of greater spending on water sanitation, interest groups devote special attention to lobbying village councils headed by women. Or it may go the other way: interest groups focus greater efforts on villages headed by men because they believe that's where they will meet the most resistance from budget makers. Whether interest group interference violated

the assumption of excludability depends on how we define the treatment effect. Interest group activity presents no threat to the exclusion restriction if we define the effect of installing a female council head to include all of the indirect repercussions that it could have on interest group activity. If, however, we seek to estimate the specific effect of having female council heads without any interference by interest groups, our experimental design may be inadequate unless we can find a way to prevent interest groups from responding strategically. These kinds of scenarios again underscore the importance of clearly stating the experimental objectives so that researchers and readers can assess the plausibility of the exclusion restriction.

2.7.2 Non-Interference

For ease of presentation, the above discussion only briefly mentioned an assumption that plays an important role in the definition and estimation of causal effects. This assumption is sometimes dubbed the Stable Unit Treatment Value Assumption, or SUTVA, but we refer to it by a more accessible name, non-interference.¹² In the notation used above, expressions such as $Y_i(d)$ are written as though the value of the potential outcome for unit i depends only upon whether or not the unit itself gets the treatment (whether d equals one or zero). A more complete notation would express a more extensive schedule of potential outcomes depending on which treatments are administered to other units. For example, for Village 1 we could write down all of the potential outcomes if only Village 1 is treated, if only Village 2 is treated, if Villages 1 and 2 are treated, and so forth. This schedule of potential outcomes quickly gets out of hand. Suppose we listed all of the potential outcomes if exactly two of the seven villages are treated: there would now be 21 potential outcomes for each village. Clearly, if our study involves just seven villages, we have no hope of saying anything meaningful about this complex array of causal effects unless we make some simplifying assumptions.

The non-interference assumption cuts through this complexity by ignoring the potential outcomes that would arise if subject i were affected by the treatment of other subjects. Formally, we reduce the schedule of potential outcomes $Y_i(d)$, where d describes all of the treatments administered to all subjects, to a much simpler schedule $Y_i(d)$, where d refers to the treatment administered to subject i .¹³ In the context of our example, non-interference implies that the sanitation budget in one village is unaffected by the gender of the council heads in other villages. Non-interference is an assumption common to both experimental and observational studies.

¹² The term "stable" in SUTVA refers to the stipulation that the potential outcomes for a given village remain stable regardless of which other villages happen to be treated. The technical aspects of this term are discussed in Rubin 1980 and Rubin 1986.

¹³ Implicit in this formulation of potential outcomes is the assumption that potential outcomes are unaffected by the overall pattern of actual or assigned treatments. In other words, $Y_i(z, d) = Y_i(z, d)$.

Is non-interference realistic in this example? It is difficult to say without more detailed information about communication between villages and the degree to which their budget allocations are interdependent. If the collection of villages were dispersed geographically, it might be plausible to assume that the gender of the village head in one village has no consequences for outcomes in other villages. On the other hand, if villages were adjacent, the presence of a woman council head in one village might encourage women in other villages to express their policy demands more forcefully. Proximal villages might also have interdependent budgets; the more one village spends on water sanitation, the less the neighboring village needs to spend in order to maintain its own water quality.

The estimation problems that interference introduces are potentially quite complicated and unpredictable. Untreated villages that are affected by the treatments that nearby villages receive no longer constitute an untreated control group. If women council heads set an example of water sanitation spending that is then copied by neighboring villages headed by men, a comparison between average outcomes in treatment villages and (semi-treated) control villages will tend to understate the average treatment effect as defined in equation (2.3), which is usually understood to refer to the contrast between treated potential outcomes and completely untreated potential outcomes. On the other hand, if female council heads cause neighboring villages headed by men to free ride on water sanitation projects and allocate less of their budget to them, the apparent difference in average budget allocations will exaggerate the average treatment effect. Given the vagaries of estimation in the face of interference, researchers often try to design experiments in ways that minimize interference between units by spreading them out temporally or geographically. Another approach, discussed at length in Chapter 8, is to design experiments in ways that allow the researcher to detect spillover between units. Instead of treating interference as a nuisance, these more complex experimental designs aim to detect evidence of communication or strategic interaction among units.

SUMMARY

In this chapter we have limited our purview to a class of randomized experiments in which treatments are deployed exactly as assigned and outcomes are observed for all of the assigned subjects. This class of studies is a natural starting point for discussing core assumptions and what they imply for research design. The chapters that follow will introduce further assumptions in order to handle the complications that arise due to noncompliance (Chapters 5 and 6) and attrition (Chapter 7).

We began by defining a causal effect as the difference between two potential outcomes, one in which a subject receives treatment and the other in which the subject does not receive treatment. The causal effect for any given subject is not directly observ-

able. However, experiments provide unbiased estimates of the average treatment effect (ATE) among all subjects when certain assumptions are met. The three assumptions invoked in this chapter are random assignment, excludability, and non-interference.

1. Random assignment: Treatments are allocated such that all units have a known probability between 0 and 1 of being placed into the treatment group. Simple random assignment or complete random assignment implies that treatment assignments are statistically independent of the subjects' potential outcomes.

This assumption is satisfied when all treatment assignments are determined by the same random procedure, such as the flip of a coin. Because random assignment may be compromised by those allocating treatments or assisting subjects, steps should be taken to minimize the role of discretion.

2. Excludability: Potential outcomes respond solely to receipt of the treatment, not to the random assignment of the treatment or any indirect by-products of random assignment. The treatment must be defined clearly so that one can assess whether subjects are exposed to the intended treatment or something else.

This assumption is jeopardized when (i) different procedures are used to measure outcomes in the treatment and control groups and (ii) research activities, other treatments, or third-party interventions other than the treatment of interest differentially affect the treatment and control groups.

3. Non-interference: Potential outcomes for observation i reflect only the treatment or control status of observation i and not the treatment or control status of other observations. No matter which subjects the random assignment allocates to treatment or control, a given subject's potential outcomes remain the same.

This assumption is jeopardized when (i) subjects are aware of the treatments that other subjects receive, (ii) treatments may be transmitted from treated to untreated subjects, or (iii) resources used to treat one set of subjects diminish resources that would otherwise be available to other subjects. See Chapter 10 for a more extensive list of examples.

Random assignment is different from the other two assumptions in that it refers to a procedure and the manner in which researchers carry it out. Excludability and non-interference, on the other hand, are substantive assumptions about the ways in which subjects respond to the allocation of treatments. When assessing excludability and non-interference in the context of a particular experiment, the first step is to carefully consider how the causal effect is defined. Do we seek to study the effect of electing women to village council positions or rather the effect of electing women from a pool of candidates that consists only of women? When defining the treatment effect of installing a female village council head, is the appropriate comparison a village with male leadership, or a male-led village with no neighboring female-led villages? Attending to these subtleties encourages a researcher to design more exacting experimental comparisons and to interpret the results with greater precision.

Attentiveness to these core assumptions also helps guide experimental investigation, urging researchers to explore the empirical consequences of different research designs. A series of experiments in a particular domain may be required before a researcher can gauge whether subjects seem to be affected by the random assignment over and above the treatment (a violation of excludability) or by the treatments administered to other units (interference).

SUGGESTED READINGS

Holland (1986) and Rubin (2008) provide non-technical introductions to potential outcomes notation. Fisher (1935) and Cox (1958) are two classic books on experimental design and analysis; Dean and Voss (1999) and Kuehl (1999) offer more modern treatments. See Rosenbaum and Rubin (1984) on the distinctive statistical properties of randomly assigned treatments.

EXERCISES: CHAPTER 2

- Potential outcomes notation:
 - Explain the notation " $Y_i(0)$."
 - Explain the notation " $Y_i(0) | D_i = 1$ " and contrast it with the notation " $Y_i(0) | d_i = 1$."
 - Contrast the meaning of " $Y_i(0)$ " with the meaning of " $Y_i(0) | D_i = 0$."
 - Contrast the meaning of " $Y_i(0) | D_i = 1$ " with the meaning of " $Y_i(0) | D_i = 0$."
 - Contrast the meaning of " $E[Y_i(0)]$ " with the meaning of " $E[Y_i(0) | D_i = 1]$."
 - Explain why the "selection bias" term in equation (2.15), $E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0]$, is zero when D_i is randomly assigned.
- Use the values depicted in Table 2.1 to illustrate that $E[Y_i(0)] - E[Y_i(1)] = E[Y_i(0) - Y_i(1)]$.
- Use the values depicted in Table 2.1 to complete the table below.
 - Fill in the number of observations in each of the nine cells.
 - Indicate the percentage of all subjects that fall into each of the nine cells. (These cells represent what is known as the joint frequency distribution of $Y_i(0)$ and $Y_i(1)$.)
 - At the bottom of the table, indicate the proportion of subjects falling into each category of $Y_i(1)$. (These cells represent what is known as the marginal distribution of $Y_i(1)$.)
 - At the right of the table, indicate the proportion of subjects falling into each category of $Y_i(0)$ (i.e., the marginal distribution of $Y_i(0)$).
 - Use the table to calculate the conditional expectation that $E[Y_i(0) | Y_i(1) > 15]$. (Hint: This expression refers to the expected value of $Y_i(0)$ given that $Y_i(1)$ is greater than 15.)
 - Use the table to calculate the conditional expectation that $E[Y_i(1) | Y_i(0) > 15]$.

$Y_i(0)$	$Y_i(1)$			Marginal distribution of $Y_i(0)$
	15	20	30	
10				
15				
20				

- Suppose that the treatment indicator d_i is either 1 (treated) or 0 (untreated). Define the average treatment effect among the treated, or ATT for short, as $\sum_1^N \tau_i d_i / \sum_1^N d_i$. Using the equations in this chapter, prove the following claim: "When treatments are allocated using complete random assignment, the ATT is, in expectation, equal to the ATE. In other words, taking expectations over all possible random assignments, $E[\tau_i | D_i = 1] = E[\tau_i]$, where τ_i is a randomly selected observation's treatment effect."
- A researcher plans to ask six subjects to donate time to an adult literacy program. Each subject will be asked to donate either 30 or 60 minutes. The researcher is considering three methods for randomizing the treatment. One method is to flip a coin before talking to each person and to ask for a 30-minute donation if the coin comes up heads or a 60-minute donation if it comes up tails. The second method is to write "30" and "60" on three playing cards each, and then shuffle the six cards. The first subject would be assigned the number on the first card, the second subject would be assigned the number on the second card, and so on. A third method is to write each number on three different slips of paper, seal the six slips into envelopes, and shuffle the six envelopes before talking to the first subject. The first subject would be assigned the first envelope, the second subject would be assigned the second envelope, and so on.
 - Discuss the strengths and weaknesses of each approach.
 - In what ways would your answer to (a) change if the number of subjects were 600 instead of 6?
 - What is the expected value of D_i (the assigned number of minutes) if the coin toss method is used? What is the expected value of D_i if the sealed envelope method is used?
- Many programs strive to help students prepare for college entrance exams, such as the SAT. In an effort to study the effectiveness of these preparatory programs, a researcher draws a random sample of students attending public high school in the United States, and compares the SAT scores of those who took a preparatory class to those who did not. Is this an experiment or an observational study? Why?
- Suppose that an experiment were performed on the villages in Table 2.1, such that two villages are allocated to the treatment group and the other five villages to the control group. Suppose that an experimenter randomly selects Villages 3 and 7 from the set of seven villages and places them into the treatment group. Table 2.1 shows that these villages have unusually high potential outcomes.
 - Define the term *unbiased estimator*.
 - Does this allocation procedure produce upwardly biased estimates? Why or why not?
 - Suppose that instead of using random assignment, the researcher placed Villages 3 and 7 into the treatment group because the treatment could be administered inexpensively in those villages. Explain why this procedure is prone to bias.
- Peisakhin and Pinto¹⁴ report the results of an experiment in India designed to test the effectiveness of a policy called the Right to Information Act (RTIA), which allows citizens to inquire about the status of a pending request from government officials. In their study, the researchers hired confederates, slum dwellers who sought to obtain ration cards (which permit the purchase of food at low cost). Applicants for such cards must fill out a

form and have their residence and income verified by a government agent. Slum dwellers widely believe that the only way to obtain a ration card is to pay a bribe. The researchers instructed the confederates to apply for ration cards in one of four ways, specified by the researchers. The control group submitted an application form at a government office; the RTIA group submitted a form and followed it up with an official Right to Information request; the NGO group submitted a letter of support from a local nongovernmental organization (NGO) along with the application form; and finally, a bribe group submitted an application and paid a small fee to a person who is known to facilitate the processing of forms.

	Bribe	RTIA	NGO	Control
Number of confederates in the study	24	23	18	21
Number of confederates who had residence verification	24	23	18	20
Median number of days to residence verification	17	37	37	37
Number of confederates who received a ration card within one year	24	20	3	5

- Interpret the apparent effects of the treatments on the proportion of applicants who have their residence verified and the speed with which verification occurred.
- Interpret the apparent effects of the treatments on the proportion of applicants who actually received a ration card.
- What do these results seem to suggest about the effectiveness of the Right to Information Act as a way of helping slum dwellers obtain ration cards?

9. A researcher wants to know how winning large sums of money in a national lottery affects people's views about the estate tax. The researcher interviews a random sample of adults and compares the attitudes of those who report winning more than \$10,000 in the lottery to those who claim to have won little or nothing. The researcher reasons that the lottery chooses winners at random, and therefore the amount that people report having won is random.

- Critically evaluate this assumption. (Hint: are the potential outcomes of those who report winning more than \$10,000 identical, in expectation, to those who report winning little or nothing?)
- Suppose the researcher were to restrict the sample to people who had played the lottery at least once during the past year. Is it now safe to assume that the potential outcomes of those who report winning more than \$10,000 are identical, in expectation, to those who report winning little or nothing?

10. Suppose researchers seek to assess the effect of receiving a free newspaper subscription on students' interest in politics. A list of student dorm rooms is drawn up and sorted randomly. Dorm rooms in the first half of the randomly sorted list receive a newspaper at their door each morning for two months; dorm rooms in the second half of the list do not receive a paper.

- University researchers are sometimes required to disclose to subjects that they are participating in an experiment. Suppose that prior to the experiment, researchers distributed a letter informing students in the treatment group that they would be

receiving a newspaper as part of a study to see if newspapers make students more interested in politics. Explain (in words and using potential outcomes notation) how this disclosure may jeopardize the excludability assumption.

- Suppose that students in the treatment group carry their newspapers to the cafeteria where they may be read by others. Explain (in words and using potential outcomes notation) how this may jeopardize the non-interference assumption.
11. Several randomized experiments have assessed the effects of drivers' training classes on the likelihood that a student will be involved in a traffic accident or receive a ticket for a moving violation.¹⁵ A complication arises because students who take drivers' training courses typically obtain their licenses faster than students who do not take a course.¹⁶ (The reason is unknown but may reflect the fact that those who take the training are better prepared for the licensing examination.) If students in the control group on average start driving much later, the proportion of students who have an accident or receive a ticket could well turn out to be higher in the treatment group. Suppose a researcher were to compare the treatment and control group in terms of the number of accidents that occur within three years of obtaining a license.
- Does this measurement approach maintain symmetry between treatment and control groups?
 - Would symmetry be maintained if the outcome measure were the number of accidents per mile of driving?
 - Suppose researchers were to measure outcomes over a period of three years starting the moment at which students were randomly assigned to be trained or not. Would this measurement strategy maintain symmetry? Are there drawbacks to this approach?
12. A researcher studying 1,000 prison inmates noticed that prisoners who spend at least three hours per day reading are less likely to have violent encounters with prison staff. The researcher therefore recommends that all prisoners be required to spend at least three hours reading each day. Let d_i be 0 when prisoners read less than three hours each day and 1 when prisoners read more than three hours each day. Let $Y_i(0)$ be each prisoner's potential number of violent encounters with prison staff when reading less than three hours per day, and let $Y_i(1)$ be each prisoner's potential number of violent encounters when reading more than three hours per day.
- In this study, nature has assigned a particular realization of d_i to each subject. When assessing this study, why might one be hesitant to assume that $E[Y_i(0) | D_i = 0] = E[Y_i(0) | D_i = 1]$ and $E[Y_i(1) | D_i = 0] = E[Y_i(1) | D_i = 1]$?
 - Suppose that researchers were to test this researcher's hypothesis by randomly assigning 10 prisoners to a treatment group. Prisoners in this group are required to go to the prison library and read in specially designated carrels for three hours each day for one week; the other prisoners, who make up the control group, go about their usual routines. Suppose, for the sake of argument, that all prisoners in the treatment group in fact read for three hours each day and that none of the prisoners

¹⁵ See Roberts and Kwan 2001.

¹⁶ Vernick et al. 1999.

in the control group read at all during the week of the study. Critically evaluate the excludability assumption as it applies to this experiment.

(c) State the assumption of non-interference as it applies to this experiment.

(d) Suppose that the results of this experiment were to indicate that the reading treatment sharply reduces violent confrontations with prison staff. How does the non-interference assumption come into play if the aim is to evaluate the effects of a policy whereby all prisoners are required to read for three hours?

CHAPTER 3

Sampling Distributions, Statistical Inference, and Hypothesis Testing

Rigorous quantification of uncertainty is a hallmark of scientific inquiry. When analyzing experimental data, the aim is not only to generate unbiased estimates of the average treatment effect but also to draw inferences about the uncertainty surrounding these estimates. Among the most attractive features of experimentation is that random allocation of treatments is a reproducible procedure. Reproducibility allows us to assess the sampling distribution, or collection of estimated ATEs that could have come about under different random assignments in order to better understand the uncertainty associated with the experiment we conducted. One objective of this chapter is to explain how experimental design affects the sampling distribution. We consider ways of designing experiments so as to reduce sampling variability, and we call attention to the fact that the sampling distribution may change markedly depending on the procedures used to randomly allocate subjects to treatment and control conditions.

A second objective is to guide the reader through the calculation and interpretation of key statistical results. When analyzing an experiment, you should consider both the estimated ATE and the uncertainty with which it is estimated. Unless you have prior information about the value of the ATE, the experimental estimate is one's best guess of the true treatment effect, but this guess may be close to or far from the true average causal effect. Statisticians commonly assess uncertainty in two ways. One method is to investigate whether the experimental results are sufficiently informative to refute a determined skeptic who insists that there is no treatment effect whatsoever. Another approach is to identify a range of values that probably bracket the true average treatment effect. This chapter introduces a flexible set of statistical techniques that may be used to assess uncertainty across a wide array of different experimental designs.

What you will learn from this chapter:

1. How to quantify the uncertainty surrounding an experimental estimate.
2. Formulas that suggest ways to design more informative experiments.
3. How to refute a determined skeptic who advances the "null hypothesis" that the treatment has no effect.
4. How to generate confidence intervals that have a 95% chance of bracketing the true sample average treatment effect.

1.1 Sampling Distributions

One of the most important topics in experimental design and analysis is sampling variability. When examining the results of a single experiment, we must bear in mind that we have in front of us just one of the many possible datasets that could have been generated via random assignment. The experiment we happened to conduct yields an estimate of the average treatment effect, but had the same observations been randomly assigned in a different way, our estimate might have been quite different. The term *sampling distribution* refers to the collection of estimates that could have been generated by every possible random assignment.¹

To illustrate the idea of a sampling distribution, let's return to the village council experiment discussed in the previous chapter. In that study, two of the seven

BOX 3.1**Definition: Sampling Distribution of Experimental Estimates**

A sampling distribution is the frequency distribution of a statistic obtained from hypothetical replications of a randomized experiment. For example, if one were to conduct the same experiment repeatedly under identical conditions, the collection of estimated average treatment effects from each replication of the experiment forms a sampling distribution. Under the central limit theorem, the sampling distribution of the estimated average treatment effect takes the shape of a normal distribution as the number of observations in treatment and control conditions increases.

¹ The distribution of estimates from all possible randomizations is also called the randomization distribution. See Rosenbaum 1984.

villages were randomly assigned to receive the treatment, which in this case is appointment of a woman to the position of village council head. Empirically, we happen to observe one particular realization of that randomization, illustrated in Table 2.2. But prior to our random allocation, there were 21 different ways to place two of the seven villages into the treatment group, and each of these 21 allocations had the same probability of being selected. Using the schedule of potential outcomes listed in Table 2.1, we may generate the hypothetical experimental results that each of the 21 possible randomizations would have produced. In other words, for each possible randomization, we calculate the average budget allocation in the treatment and control groups, and calculate the difference in means. The results are displayed in Table 3.1.

The first thing to note about the results in Table 3.1 is that the average estimated ATE is 5, which is exactly the same as the true ATE in Table 2.1. This is no coincidence. In fact, the exercise underscores a very important feature of randomized experiments: given three core assumptions discussed in Chapter 2 (random assignment, excludability, and non-interference), the average estimated ATE across all

TABLE 3.1

Sampling distribution of estimated ATEs generated when two of the seven villages listed in Table 2.1 are assigned to treatment

Estimated ATE	Frequency with which an estimate occurs
-1	2
0	2
0.5	1
1	2
1.5	2
2.5	1
6.5	1
7.5	3
8.5	3
9	1
9.5	1
10	1
16	1
Average	5
Total	21

ossible random assignments is equal to the true ATE. Any single experiment might give a number that is too high or too low, but the expected value of this estimation procedure is the true ATE. One of the great virtues of experiments is that they generate unbiased estimates of the ATE: merely by subtracting the control group mean from the treatment group mean, we obtain an estimator that on average recovers the true ATE.

The next thing to note about the results in Table 3.1 is that the 21 possible experiments generate quite different results. The largest estimated ATE is 16—we had a 1-in-21 chance of obtaining an estimate that was more than three times the size of the true ATE of 5. Two of the 21 randomizations produce an estimated ATE of -1 . Had we obtained an estimate of -1 , we might have been led to believe that women village council heads tend to reduce the share of budgets directed at water sanitation, even though the opposite is true. The dispersion of estimates around the true ATE reminds us that while experiments are unbiased, they are not necessarily precise. With just seven observations, our experiment generates results that vary markedly from one randomization to the next.

3.2 The Standard Error as a Measure of Uncertainty

In order to describe the precision with which an experiment recovers the ATE, we need a statistic that characterizes the amount of sampling variability. Sampling variability is typically expressed by reference to the *standard error*. The larger the standard error, the more uncertainty surrounds our parameter estimate. The standard error is the standard deviation of the sampling distribution. It is obtained by calculating the squared deviation of each estimate from the average estimate, dividing by the number of possible randomizations, and taking the square root of the result.

Based on the numbers in Table 3.1, we calculate the standard error as follows:

Sum of squared deviations

$$\begin{aligned} &= (-1 - 5)^2 + (-1 - 5)^2 + (0 - 5)^2 + (0 - 5)^2 + (0.5 - 5)^2 + (1 - 5)^2 \\ &+ (1 - 5)^2 + (1.5 - 5)^2 + (1.5 - 5)^2 + (2.5 - 5)^2 + (6.5 - 5)^2 + (7.5 - 5)^2 \\ &+ (7.5 - 5)^2 + (7.5 - 5)^2 + (8.5 - 5)^2 + (8.5 - 5)^2 + (8.5 - 5)^2 + (9 - 5)^2 \\ &+ (9.5 - 5)^2 + (10 - 5)^2 + (16 - 5)^2 = 445. \end{aligned}$$

$$\text{Square root of the average squared deviation} = \sqrt{\frac{1}{21}(445)} = 4.60. \quad (3.1)$$

When a parameter is estimated using an unbiased estimator (such as difference-in-means), a helpful rule of thumb is that approximately 95% of the experimental outcomes fall within an interval that ranges from two standard errors below the true

BOX 3.2

Definition: Standard Deviation

The standard deviation of a variable X is

$$\sqrt{\frac{1}{N} \sum_1^N (X_i - \bar{X})^2},$$

where $\bar{X}$ denotes the mean of X . Notice that this formula divides by N . When X is a random sample from a larger population containing N^* subjects whose mean is unknown, the estimate of the population standard deviation is

$$\sqrt{\frac{1}{N-1} \sum_1^N (X_i - \bar{X})^2}.$$

Definition: Standard Error

The standard error is the standard deviation of a sampling distribution. Suppose there are J possible ways of randomly assigning subjects. Let $\hat{\theta}_j$ represent the estimate (denoted by the “hat” mark: $\hat{}$) we obtain from the j th randomization, and $\bar{\theta}$ represent the average estimate for all J . For example, $\hat{\theta}_j$ may represent the difference-in-means estimate from one of the possible random assignments. Over all J possible random assignments, the standard error of $\hat{\theta}$ is

$$\sqrt{\frac{1}{J} \sum_1^J (\hat{\theta}_j - \bar{\theta})^2}.$$

parameter to two standard errors above it. Given a true parameter of 5 and a standard error of 4.60, this interval stretches from -4.20 to 14.20 . (In Table 3.1, we see that in fact this rule of thumb works well for this example: 20 of the 21 estimates, or 95%, fall in this range. The approximation on which this rule of thumb rests tends to become more accurate as sample size increases.) At the low end of this interval, we infer that appointing a female council head diminishes budgets for water sanitation, and at the top of this interval we grossly exaggerate the extent to which women leaders allocate more money toward this policy area.

How can we reduce the standard error? In other words, how can we design our experiment so that it produces more precise estimates of the ATE? In order to answer this question, let's inspect a formula that expresses the standard error as a function of the potential outcomes and the experimental design. Before getting to the formula itself, we first define some key terms. A *variance* of an observed or potential outcome

BOX 3.3

Rule of Thumb for Sampling Distributions of the Estimated ATE

For randomized experiments with a given standard error, approximately 95% of all random assignments will generate estimates of the ATE that fall within ± 2 standard errors from the true ATE. For example, if the standard error is 10 and the ATE is 50, approximately 95% of the estimates will be between 30 and 70. This rule of thumb works best when N is large.

For a set of N subjects, the average squared deviation of each subject's value from the mean for all N subjects. For example, the variance of $Y_i(1)$ is:

$$\text{Var}(Y_i(1)) = \frac{1}{N} \sum_1^N \left(Y_i(1) - \frac{\sum_1^N Y_i(1)}{N} \right)^2 \quad (3.2)$$

The higher the variance, the greater the dispersion around the mean. The smallest possible variance is zero, which implies that the variable is a constant. Note that the variance is the square of the standard deviation. For more details on this formula, see Box 3.2.

To obtain the covariance between two variables, such as $Y_i(1)$ and $Y_i(0)$, subtract the mean from each variable, and then calculate the average cross-product of the result:

$$\text{Cov}(Y_i(0), Y_i(1)) = \frac{1}{N} \sum_1^N \left(Y_i(0) - \frac{\sum_1^N Y_i(0)}{N} \right) \left(Y_i(1) - \frac{\sum_1^N Y_i(1)}{N} \right) \quad (3.3)$$

The covariance is a measure of association between two variables. A negative covariance implies that lower values of one variable tend to coincide with higher values of the other variable. A positive covariance means that higher values of one variable tend to coincide with higher values of the other variable.

Applying these formulas to the schedule of potential outcomes listed in Table 2.1, we find the variance of $Y_i(0)$ to be 14.29, and the variance of $Y_i(1)$ to be 42.86. The covariance of $Y_i(0)$ and $Y_i(1)$ is 7.14. In order to obtain the standard error associated with the experimental estimate of the average treatment effect, the remaining step is to decide how many of the N observations are to be treated. We will call m the number of treated units. In our example, $m = 2$ and $N = 7$. In general, we require that $0 < m < N$, because if m were zero or equal to N , we would have just one experimental group rather than two. For the same reason, we also require $N > 1$.

With all of the ingredients in place, we are now ready to write the formula for the standard error of the estimated ATE. The equation places a "hat" mark over the estimand in order to indicate that we are talking about an estimate of the ATE, not the true ATE:

BOX 3.4

Properties of Variances and Covariances

- (a) $\text{Cov}(Y_i(0), Y_i(0)) = \text{Var}(Y_i(0)) \geq 0$
- (b) $\text{Cov}(Y_i(0), Y_i(1)) = \text{Cov}(Y_i(0), Y_i(0) + \tau_i)$
 $= \text{Var}(Y_i(0)) + \text{Cov}(Y_i(0), \tau_i)$
- (c) $\text{Cov}(aY_i(0), bY_i(1)) = ab\text{Cov}(Y_i(0), Y_i(1))$

Covariances are bounded by the restriction that:

$$\text{Cov}(Y_i(0), Y_i(1)) \leq \sqrt{\text{Var}(Y_i(0))\text{Var}(Y_i(1))} \leq \frac{\text{Var}(Y_i(0)) + \text{Var}(Y_i(1))}{2}$$

The correlation between two variables $Y_i(0)$ and $Y_i(1)$ is

$$\frac{\text{Cov}(Y_i(0), Y_i(1))}{\sqrt{\text{Var}(Y_i(0))\text{Var}(Y_i(1))}}$$

$$SE(\widehat{ATE}) = \sqrt{\frac{1}{N-1} \left\{ \frac{m\text{Var}(Y_i(0))}{N-m} + \frac{(N-m)\text{Var}(Y_i(1))}{m} + 2\text{Cov}(Y_i(0), Y_i(1)) \right\}} \quad (3.4)$$

This formula tells us which factors reduce the size of the standard error.² The formula contains five inputs: N , m , $\text{Var}(Y_i(0))$, $\text{Var}(Y_i(1))$, and $\text{Cov}(Y_i(0), Y_i(1))$. Changing each input one by one, while holding the other inputs constant, we can examine how the standard error changes. Here is a summary of the formula's implications for experimental design:

1. The larger the N , the smaller the standard error. Holding constant the other inputs (including m , the size of the treatment group) and increasing N means increasing the control group. As the control group grows, the first and third terms inside the braces of equation (3.4) are diminished by an expanding $N - 1$ denominator. If the control group were infinite in size, the only source of uncertainty would come from the treatment group. Sometimes adding subjects to the control group involves little or no additional cost. For experiments in which treating subjects costs resources but leaving them untreated does not (e.g., sending mail to those in the treatment group while sending nothing to those in the control), bring as many additional subjects as possible into the control group. A similar point holds

² Equation (3.4) describes the true standard error, $SE(\widehat{ATE})$, which is not to be confused with the estimated standard error, $\widehat{SE}(\widehat{ATE})$, calculated based on a particular experiment.

for the treatment group: holding the control group's size constant, increasing m , the size of the treatment group, diminishes the second and third terms in equation (3.4). Of course, where possible, increase the size of both the control group and the treatment group, as this reduces all three terms in equation (3.4).

The smaller the variance of $Y_i(0)$ or $Y_i(1)$, the smaller the standard error. To maximize precision, conduct experiments on observations that are as similar as possible in terms of their potential outcomes. This principle has three design implications. First, it encourages researchers to measure outcomes as accurately as possible, as this dampens variability. Second, as discussed later in this chapter, blocking may be used to improve precision by grouping observations with similar potential outcomes.³ Third, as explained in Chapter 4, another way to reduce variance in outcomes is to measure outcomes in advance of the experimental intervention, sometimes known as a *pre-test*, in addition to measuring outcomes after the intervention via a *post-test*. Instead of defining the experimental outcome to be the score on the post-test, define the experimental outcome to be the change from pre-test to post-test. Change scores usually have less variance than post-test scores.

4. Assuming that $Y_i(0)$ and $Y_i(1)$ do vary, the smaller the covariance between $Y_i(0)$ and $Y_i(1)$, the smaller the standard error.⁴ A particularly favorable case occurs when the potential outcomes have negative covariance: that is, where high values of $Y_i(0)$ tend to coincide with low values of $Y_i(1)$. In order to see how this pattern might occur, write $Y_i(1) = Y_i(0) + \tau_i$. Substituting for $Y_i(1)$ allows us to express $\text{Cov}(Y_i(0), Y_i(1))$ as $\text{Var}(Y_i(0)) + \text{Cov}(Y_i(0), \tau_i)$. (See Box 3.4.) So, when $Y_i(0)$ and τ_i are negatively related (e.g., students with low baseline scores are most helped by the treatment), the covariance between $Y_i(0)$ and $Y_i(1)$ may be close to zero or even negative.

1. A subtle implication of the formula is that when the variances of $Y_i(0)$ and $Y_i(1)$ are similar, it is advisable to assign approximately half of the observations to the treatment group, such that $m \approx N/2$. When the potential outcomes have different variances, invest additional observations to the experimental condition with

greater variance. For example, if $\text{Var}(Y_i(1)) > \text{Var}(Y_i(0))$, put a greater share of the observations into the treatment group. In practice, however, researchers seldom know in advance which group is likely to have more variance and therefore place equal numbers of subjects in each condition.

In order to see the formula at work, we fill in the values obtained from Table 2.1 and obtain exactly the same number we calculated from Table 3.1:

$$SE(\widehat{ATE}) = \sqrt{\frac{1}{6} \left\{ \frac{(2)(14.29)}{5} + \frac{(5)(42.86)}{2} + (2)(7.14) \right\}} = 4.60. \quad (3.5)$$

When we know the full schedule of potential outcomes, equation (3.4) tells us the standard deviation of estimated ATEs from all possible random assignments.

Equation (3.4) can be used to demonstrate that our design of $m = 2$ is less than optimal. Increasing the number of treated units so that $m = 3$ lowers the standard error to 3.7. Raising m to 4 lowers the standard error even further, to 3.3. The reason it is better to put more observations into the treatment group than the control group is that in this example the treated potential outcomes have more variance than the untreated potential outcomes: $\text{Var}(Y_i(1)) > \text{Var}(Y_i(0))$. Raising m to 5 goes too far in that direction and leads to a slight deterioration in precision.

To summarize, standard errors are measures of uncertainty; they indicate the extent to which estimates will vary across all possible random assignments. One consideration when designing experiments is to keep standard errors small. Two types of inputs determine the standard error: the schedule of potential outcomes and the number of observations assigned to treatment and control groups. The schedule of potential outcomes is unobserved, but experimenters sometimes have opportunities to design experiments in ways that limit the mischief that highly variable potential outcomes may cause. Variance in $Y_i(0)$ and $Y_i(1)$ may be reduced by measuring outcomes in a more precise manner or by blocking observations into homogeneous subsets. Standard errors are also reduced by "Robin Hood treatments" that raise outcomes among those with low $Y_i(0)$ while lowering outcomes among those with high $Y_i(0)$. (For example, a regimented physical fitness program in a military academy, where cadets have little spare time, might raise the level of fitness among the least fit while lowering it among top-level athletes.) If the cost per subject is similar in both experimental groups, assign similar numbers to treatment and control, tilting the balance in favor of the group that is expected to have more variable outcomes.

3.3 Estimating Sampling Variability

The previous section illustrated the concept of sampling variability by showing how standard errors can be calculated from a known schedule of potential outcomes.

The researcher could restrict attention to subjects with similar background attributes in order to reduce variance in $Y_i(0)$ and $Y_i(1)$, but this approach has the drawback of limiting the generalizations that might be drawn from this narrow set of subjects. In order to overcome this limitation, a researcher may conduct the experiment within several different blocks, each of which contains subjects with similar background attributes.

Here is the underlying intuition for why positive covariance leads to larger standard errors. If high values of $Y_i(1)$ tend to coincide with high values of $Y_i(0)$, then selecting a subject with a high potential outcome into the treatment group leaves one fewer subject with high potential outcomes for the control group. Positive covariance between $Y_i(0)$ and $Y_i(1)$ therefore means that results are sensitive to the placement of subjects into treatment or control. On the other hand, if high values of $Y_i(1)$ tend to coincide with low values of $Y_i(0)$, then selecting a subject with high potential outcomes into the treatment group leaves one fewer subject with low potential outcomes for the control group. In this case, there is less sampling variability because the control group is "compensated" for the fact that the treatment group received a high

certainty can be addressed through experimental design. Now that we have an appreciation for what a standard error is, we take up the question that arises when one estimates an ATE for a particular set of subjects.⁵ Suppose the researcher wants to know the standard error in order to calibrate the uncertainty associated with his estimate. The researcher has neither the complete schedule of potential outcomes nor results from all of the hypothetical random assignments that could have allocated this set of observations to treatment and control groups. Instead, this researcher has results from a single randomization.

Assessing uncertainty is an estimation problem. The true standard error is unknown. We seek to estimate this unknown quantity using data from a single experiment. What hints can a single experiment provide about how other experimental assignments might have come out? Returning to equation (3.4), we see that the experiment provides information about four of the five inputs that generate standard errors. The number of observations allocated to treatment and control is known. The variance of outcomes in the untreated potential outcomes $Y_i(0)$ can be estimated using the observed outcomes in the control group. Because the control group is assigned at random, the variance that we observe in the control group is an unbiased estimate of the variance in $Y_i(0)$. The same approach can be used to estimate the variance of $Y_i(1)$ based on the observed variance in the assigned treatment group.

The one element in this equation that cannot be estimated empirically is the covariance between $Y_i(0)$ and $Y_i(1)$, as we never observe both potential outcomes for the same subject. The standard approach is to use a conservative estimation formula that is at least as large as equation (3.4) regardless of the covariance between $Y_i(0)$ and $Y_i(1)$.⁶ The conservative formula assumes that the treatment effect is the same for all subjects, which implies that the correlation between $Y_i(0)$ and $Y_i(1)$ is 1.0.⁷

⁵ The average treatment effect is the ATE for the "finite population" of subjects in one's experiment. If the experimental subjects are seen as a sample from a larger population, one may distinguish between the ATE in the sample and the ATE in the population. In Chapter 11, we take up the question of how to use the sample at hand to draw inferences about the ATE in the population.

⁶ To be more precise, the conservative estimation approach tends to overestimate the true sampling variance, which is the square of the standard error depicted in equation (3.4). One note of caution: although the conservative estimation formula overestimates the true sampling variance on average, estimation is subject to sampling variability. A given estimate of the sampling variance using the conservative formula may still be smaller than the true sampling variance.

⁷ See Freedman, Pisani, and Purves 1998, pp. A32–A34. The conservative formula will give unbiased estimates of the sampling variance under either of two scenarios. The first arises when the treatment effect τ_i is the same for all subjects. The second arises when subjects are sampled at random from a large population prior to random assignment, and the objective is to estimate the population average treatment effect. When subjects are sampled randomly from a large population, the selection of one subject for the treatment group has no material effect on the pool of available subjects that can be selected into the control group, which renders the covariance between $Y_i(0)$ and $Y_i(1)$ irrelevant.

The formula for estimating the standard error of the average treatment effect is

$$\widehat{SE} = \sqrt{\frac{\widehat{\text{Var}}(Y_i(0))}{N-m} + \frac{\widehat{\text{Var}}(Y_i(1))}{m}}, \quad (3.6)$$

where the variances are estimated using the m observations of potential outcomes from the treatment group:

$$\widehat{\text{Var}}(Y_i(1)) = \frac{1}{m-1} \sum_1^m \left(Y_i(1) | d_i = 1 - \frac{\sum_1^m Y_i(1) | d_i = 1}{m} \right)^2 \quad (3.7)$$

and the $N - m$ observations of potential outcomes from the control group:

$$\widehat{\text{Var}}(Y_i(0)) = \frac{1}{N-m-1} \sum_{m+1}^N \left(Y_i(0) | d_i = 0 - \frac{\sum_{m+1}^N Y_i(0) | d_i = 0}{N-m} \right)^2. \quad (3.8)$$

Note that when calculating the sample variances in equations (3.7) and (3.8), we divide by one less than the number of observations to take into account the fact that one observation is expended when we calculate the sample mean.⁸ In order to avoid dividing by zero when estimating the standard error, we must have at least two subjects in each experimental group.

How closely do the empirical estimates⁹ of the standard errors match the true standard errors? Using the conservative formula, the average standard error in our example is 4.65, which is not far from the true standard error of 4.60. Although the estimates vary from one random allocation to the next, on average, the estimator depicted in equation (3.6) does a reasonably good job of estimating the true level of sampling variability.

3.4 Hypothesis Testing

The previous section illustrated the challenges that arise when estimating the sampling uncertainty surrounding a single experiment's estimate of the ATE. Accurate estimation of the standard error requires accurate guesses about the variances and covariance of potential outcomes. If we do not have much data about the variances or if our simplifying assumption about the unobservable covariance is mistaken, our estimated standard errors may be inaccurate.

⁸ If we know the mean and the outcomes for all but one of the subjects, we can deduce the outcome for the remaining subject. In statistical parlance, calculating the mean expends one *degree of freedom*.

⁹ The 21 estimated standard errors are {1.581, 1.871, 1.871, 1.871, 1.871, 1.871, 2.236, 2.784, 2.958, 3.122, 3.122, 5.244, 5.339, 7.599, 7.665, 7.730, 7.730, 7.730, 7.730, 7.826, 7.826}.

Much easier is the task of testing the *sharp null hypothesis* that the treatment effect is zero for all observations. What makes it easy is that if the null hypothesis is true, $Y_i(0) = Y_i(1)$. Under this special case, we observe *both* potential outcomes for every observation. We may therefore take the observed outcomes in our dataset and impute all possible randomizations as though we were working from a complete schedule of potential outcomes. These simulated randomizations provide an exact sampling distribution of the estimated average treatment effect under the sharp null hypothesis. By looking at this distribution of hypothetical outcomes, we can calculate the probability of obtaining an estimated ATE at least as large as the one we obtained from our actual experiment if the treatment effect were in fact zero for every subject.

For example, the randomization depicted in Table 2.2 generated an estimate of the ATE of 6.5. How likely are we to obtain an estimate as large as or larger than 6.5 if the true effect were zero for all observations? The probability, or *p*-value, of interest in this case addresses a *one-tailed hypothesis*, namely that female village council heads increase budget allocations to water sanitation. If we sought to evaluate the *two-tailed hypothesis*—whether female village council heads either increase or decrease the budget allocation for water sanitation—we would calculate the *p*-value of obtaining a number that is greater than or equal to 6.5 or less than or equal to -6.5 .

Based on the observed outcomes in Table 2.2, we may calculate the 21 possible estimates of the ATE that could have been generated if the null hypothesis were true: $-7.5, -7.5, -7.5, -4.0, -4.0, -4.0, -4.0, -4.0, -0.5, -0.5, -0.5, -0.5, -0.5, -0.5, 0.0, 3.0, 6.5, 6.5, 6.5, 10.0, 10.0$. Five of the estimates are as large as 6.5. So when evaluating the one-tailed hypothesis that female village heads *increase* water sanitation budgets, we would conclude that the probability of obtaining an estimate as large as 6.5 if the null hypothesis were true is $5/21 = 24\%$. A two-tailed hypothesis test would count all instances in which the estimates are at least as great as 6.5 in *absolute value*. Eight of the estimates qualify, so the two-tailed *p*-value is $8/21 = 38\%$.

In theory, this type of calculation could be performed on experiments of any size, but in practice the number of possible random assignments becomes astronomical as N increases. For example, an experiment where $N = 50$ and half the observa-

tions are assigned to the treatment group potentially generates more than 126 trillion randomizations:

$$\frac{50!}{25!25!} = 126,410,606,437,752. \quad (3.9)$$

When the number of possible randomizations is large, one can closely approximate the sampling distribution by sampling at random from the set of all possible random assignments. Whether one uses all possible randomizations or a large sample of them, the calculation of *p*-values based on an inventory of possible randomizations is called *randomization inference*.

This approach to hypothesis testing has two attractive properties. First, the procedures used to calculate *p*-values may be applied to a very broad class of hypotheses and applications. The method is not confined to large samples or normally distributed outcomes. Any sample size will do, and the method can be applied to all sorts of outcomes, including counts, durations, or ranks. Second, the method is exact in the sense that the set of all possible random assignments fully describes the sampling distribution under the null hypothesis. By contrast, the hypothesis testing methods discussed in introductory statistics courses rely on an approximation in order to derive the shape of the sampling distribution. (See Box 3.7.) Although exact and approximate methods will tend to give very similar answers when samples are large, we use randomization inference throughout the book so that a single statistical approach may be applied to a broad array of applications without approximations or additional assumptions.

Since the 1930s, it has been conventional to dub *p*-values that are below 0.05 as *statistically significant* on the grounds that, under the null hypothesis, the researcher had less than a 1-in-20 chance of obtaining the result due to chance. The *p*-values in the village council experiment were 0.24 and 0.38, which fail to meet this standard. By the 0.05 standard of statistical significance, the estimate of 6.5 does not provide a convincing refutation of the null hypothesis of no effect. The 0.05 standard is a matter of convention, not statistical theory, but it is so deeply entrenched that researchers should be prepared to indicate whether their experimental results are significant at the 0.05 level.

Anticipating this concern, researchers often attempt to forecast the probability that their experiment, when conducted, will lead to the rejection of the null hypothesis at a given significance level, such as 0.05. This probability is termed the *statistical power* of the experiment. To say that an experimental design has 80% power, for instance, means that 80% of all possible random assignments will produce observed results that will lead to the rejection of the null hypothesis in the presence of the posited treatment effect. Forecasting the power of an experiment requires some guesswork. The appendix to this chapter illustrates how assumptions are used when calculating the power of a proposed experiment.

Be careful not to confuse statistical significance with substantive significance. A parameter estimate that falls short of the 0.05 threshold might nevertheless be

BOX 3.5

Definition: Sharp Null Hypothesis of No Effect

The treatment effect is zero for all subjects. Formally, $Y_i(1) = Y_i(0)$ for all i .

Definition: Null Hypothesis of No Average Effect

The average treatment effect is zero. Formally, $\mu_{Y(1)} = \mu_{Y(0)}$.

BOX 3.6**Definition: One-tailed and Two-tailed Hypothesis Tests**

A null hypothesis that specifies that the treatment effect is zero can be rejected by test statistics that are either very large or very small. This is called a two-tailed test. For example, an intervention that is believed to *change* outcomes (either positively or negatively) would be evaluated using a two-tailed test. A null hypothesis that specifies that the effect is zero or less can be rejected by a large positive test statistic. (Similarly, a null hypothesis that specifies that an effect is zero or more is rejected by a large negative test statistic.) This is called a one-tailed test. Therapeutic interventions are advanced in anticipation of finding positive effects, in which case the null hypothesis is that the effect is zero or less, and an appropriate test is one-tailed.

Definition: p -value

For a two-tailed test, the p -value is the probability of obtaining a test statistic at least as large in absolute value as the observed test statistic, given that the null hypothesis is true. For example, suppose the null hypothesis is that the treatment has no effect, and the test statistic is the estimated ATE. If the estimated ATE is 5 and the p -value is 0.20, there is a 20% chance of obtaining an estimate as large as 5 (or as small as -5) simply by chance. If the alternative hypothesis is a positive (or negative) effect, a one-tailed test is used.

Randomization Inference

The sampling distribution of the test statistic under the null hypothesis is computed by simulating all possible random assignments. When the number of random assignments is too large to simulate, the sampling distribution may be approximated by a large random sample of possible assignments. p -values are calculated by comparing the observed test statistic to the distribution of test statistics under the null hypothesis.

important and interesting. If female council heads in fact caused a 6.5 percentage-point increase in budgetary allocations to water sanitation, the health consequences of increasing the role of women in local government throughout the Indian countryside could be profound. Although statistical uncertainty remains, the data have taught us something potentially useful. If this were the first experiment of its kind and we had no prior knowledge of the treatment effect, the estimate of 6.5 would still be our best guess of the true ATE, despite the fact that we cannot rule out the

BOX 3.7**Comparing Randomization Inference to T-Tests**

Approximate methods for testing the sharp null hypothesis of no effect assume that the sampling distribution of the difference-in-means estimator has a particular shape. For example, the t -test, which should be familiar to those who have taken an introductory statistics course, assumes that the sampling distribution of the estimated difference-in-means follows a t -distribution, which is similar to a normal distribution. The t -test gives accurate p -values when outcomes in each of the experimental groups are distributed normally. When outcomes are distributed non-normally, approximate methods become increasingly accurate as the number of subjects in each of the experimental conditions grows.

To illustrate the difference between a t -test and randomization inference, we generated the hypothetical dataset shown below. The treatment and control groups each contain 10 subjects. The treatment is an encouragement to make a charitable donation, and the outcome is the amount of money contributed. This outcome is skewed to the right due to a few large donations. The treatment group average is 80, while the control group average is 10. The sampling distribution under the sharp null hypothesis is bimodal (due to the influence of a few large donations), but conventional methods assume that the sampling distribution is bell-shaped. Repeating the random assignment 100,000 times under the sharp null hypothesis of no effect, we find that the observed average treatment effect of 70 has a one-tailed p -value of 0.032. A t -test assuming equal variances puts this p -value at 0.082; a t -test allowing for unequal variances declares the p -value to be 0.091. The t -test is inaccurate in this case because the outcomes are skewed and the number of observations is small. Readers are encouraged to tinker with this example using the simulation code at <http://isps.research.yale.edu/FEDAI> to see how the shape of the sampling distribution changes as more subjects are added to the dataset.

Treatment	Donation
1	500
1	100
1	100
1	50
1	25
1	25
1	0

Treatment	Donation
1	0
1	0
1	0
0	25
0	20
0	15
0	15

Treatment	Donation
0	10
0	5
0	5
0	5
0	0
0	0

pothesis that the ATE is zero. The right way to think about an experimental result that is substantively significant but statistically insignificant is that it warrants further investigation. As we conduct further experiments, our uncertainty will gradually diminish, and we will be able to make a clearer determination about the true value of the ATE. (See Chapter 11 for a discussion of how sampling variability diminishes when pooling a series of repeated experiments.)

Conversely, don't be overly impressed with statistically significant results without reflecting on their substantive significance. All else equal, the standard error declines proportionally to the square root of N . Imagine the number of observations were to grow from 7 to 70,000 villages. This 10,000-fold increase in N would imply a 100-fold decrease in standard error, from 4.6 to 0.046 percentage-points. If this enormous experiment were to indicate that female village council heads increased sanitation budgets by an average of 0.1 percentage points, the finding would be statistically significant at the 0.05 level but substantively trivial.

One final caution about testing the null hypothesis that $Y_i(0) = Y_i(1)$: this equality represents the sharp null hypothesis of *no treatment effect for any observation*, which should not be confused with the null hypothesis of *no average effect*. Suppose the treatment effect were 5 for half of the observations and -5 for the other half. In this scenario, the average effect would be zero, but the sharp null hypothesis would be false. Both null hypotheses have the testable implication that the average outcome in the treatment group will be similar to the average outcome in the control group, but the sharp null hypothesis has the further testable implication that the two groups will have similarly shaped distributions.

In order to conduct a more exacting evaluation of the sharp null hypothesis, we may consider statistics other than the estimated ATE. Because the sharp null hypothesis provides us with the complete schedule of potential outcomes, we can simulate the sampling distribution of any test statistic. For example, we could compare the variances in the treatment and control groups. If treatment effects vary and are uncorrelated with or positively correlated with $Y_i(0)$, the observed variance of $Y_i(1)$ will tend to be larger than the variance of $Y_i(0)$. The sampling distribution under the sharp null hypothesis indicates the probability of observing a difference in variances as large as what we in fact observe in our sample. We simply calculate all possible differences and compute the p -value of the difference we obtained from our experiment. We return to these and other diagnostic tests of heterogeneous treatment effects in Chapter 9.

3.5 Confidence Intervals

When faced with a decision, policy makers may turn to an experiment for guidance about the average effect of an intervention. They typically want to know how big the

zero. Testing the null hypothesis is beside the point; their principal objective is to use the results to form a guess about the value of the ATE.

Interval estimation is a statistical procedure that uses the data to generate a probability statement about the range of values within which a parameter is located. For example, suppose a researcher conducts an experiment in an attempt to learn the value of the ATE. Using a formula that will be described momentarily, the researcher generates a 95% confidence interval that ranges from (2, 10). This interval has a 0.95 probability of bracketing the true ATE. In other words, if we imagine a series of hypothetical replications of this experiment under identical conditions, 95 out of 100 random assignments will generate intervals that bracket the true ATE. When interpreting confidence intervals, remember that the location of the interval varies from one experiment to the next, while the true ATE remains constant.

By convention, social scientists construct 95% confidence intervals, but forming a 95% confidence interval based on the results of a single experiment requires some guesswork. Recall that a single experiment reveals the $Y_i(1)$ outcomes for the treated subjects and $Y_i(0)$ outcomes for the untreated subjects. We do not observe the full schedule of potential outcomes. Without the full schedule of potential outcomes, we cannot simulate the sampling distribution of the estimated ATE. The best we can do is approximate the full schedule of potential outcomes by making an educated guess about the unobserved $Y_i(0)$ outcomes for the treated subjects and $Y_i(1)$ outcomes for the untreated subjects.

The most straightforward method for filling in missing potential outcomes is to assume that the treatment effect τ_i is the same for all subjects. For subjects in the control condition, missing $Y_i(1)$ values are imputed by adding the estimated ATE to the observed values of $Y_i(0)$. Similarly, for subjects in the treatment condition, missing $Y_i(0)$ values are imputed by subtracting the estimated ATE from the observed values of $Y_i(1)$. This approach yields a complete schedule of potential outcomes, which we may then use to simulate all possible random allocations. In order to form a 95% confidence interval, we list the estimated ATE from each random allocation in ascending order. The estimate at the 2.5th percentile marks the bottom of the interval, and the estimate at the 97.5th percentile marks the top of the interval.¹⁰

When the treatment and control groups contain an equal number of subjects ($m = N - m$), this method of forming confidence intervals will tend to be conservative, in the sense that the interval typically will be wider than the interval we

¹⁰ A more complex and computationally intensive approach is to "invert" the hypothesis test (Rosenbaum 2002). This method involves hypothesizing an ATE called ATE^* , subtracting it from the observed outcomes in the treatment group to approximate $Y_i(0)$, and testing the null hypothesis that the average (adjusted) $Y_i(0)$ in the treatment group is equal to the average $Y_i(0)$ in the control group. Progressively larger values of ATE^* are tried until one locates a value that has a p -value of 0.025 and another that has a p -value of 0.975; these values mark the range of the 95% confidence interval. This method tends to be more accurate than the simpler method we describe, but both tend to produce similar results, especially

could obtain if we actually observed $Y_i(0)$ and $Y_i(1)$ for all subjects.¹¹ When the number of subjects in treatment and control differ, this method may produce confidence intervals that are too wide or too narrow, depending on whether $\text{Var}(Y_i(0))$ is larger or smaller than $\text{Var}(Y_i(1))$. The method is conservative when the control group is larger and $\text{Var}(Y_i(0)) > \text{Var}(Y_i(1))$ or when the treatment group is larger and $\text{Var}(Y_i(1)) > \text{Var}(Y_i(0))$. When working with treatment and control groups of markedly different sizes, one should use this method for estimating confidence intervals with caution if outcomes in the smaller group are substantially more variable. A rule of thumb is that caution is required if the smaller group is less than half the size of the larger group, and the smaller group's standard deviation is at least twice as large as the larger group's standard deviation. One reason to assign equal numbers of subjects to treatment and control is that this type of "balanced" experimental design facilitates interval estimation by eliminating these complications.

To illustrate interval estimation, we analyze data from Clingingsmith, Khwaja, and Kremer's study of Pakistani Muslims who participated in a lottery to obtain a visa for the pilgrimage to Mecca.¹² By comparing lottery winners to lottery losers, the authors are able to estimate the effects of the pilgrimage¹³ on the social, religious, and political views of the participants. Here, we consider the effect of winning the visa lottery on attitudes toward people from other countries. Winners and losers were asked to rate the Saudi, Indonesian, Turkish, African, European, and Chinese people on a five-point scale ranging from very negative (-2) to very positive ($+2$). Adding the responses to all six items creates an index ranging from -12 to $+12$.

The distribution of responses in the treatment group ($N = 510$) and control group ($N = 448$) is presented in Table 3.2. The average in the treatment group is 2.34, as compared to 1.87 in the control group. The estimated ATE is therefore 0.47. The estimated standard deviations in the treatment and control groups are 2.63 and 2.41, respectively. In this study, the treatment and control conditions contain similar numbers of subjects, and the smaller of the two groups does not have more vari-

11 In small samples, the distance between the estimated ATE and the top (or bottom) of the estimated confidence interval should be widened by a factor of $\sqrt{(N-1)/(N-2)}$ to adjust for the fact that the ATE used to fill in the full schedule of potential outcomes is itself estimated from the data. For example, suppose a study of ten subjects were to produce an estimated interval ranging from 8 to 12. This correction would widen the interval by a factor of 1.061, resulting in an adjusted interval that ranges from 7.88 to 12.12. See Samit and Aronow 2011. As N increases, this correction becomes negligible.

12 Clingingsmith, Khwaja, and Kremer 2009.

13 Our description sidesteps the complications that arise due to the fact that 14% of the lottery losers nonetheless made the pilgrimage and 1% of the lottery winners failed to make the pilgrimage. Anticipating the discussion of noncompliance in Chapters 5 and 6, we estimate the intent-to-treat effect: the effect of winning the lottery. In order to simplify the presentation, we sidestep the problem of clustering (multiple individuals applying for the lottery together) by randomly selecting one person from each cluster. For ease of presentation, we also ignore the slight differences in treatment probabilities across blocks defined by the size and location of the parties applying for a visa. Controlling for blocks has negligible effects on the results.

TABLE 3.2

Pakistani Muslims' ratings of peoples from foreign countries by success in the visa lottery

Ratings of people from other countries	Distribution of responses	
	Control [%]	Treatment [%]
-12	0.00	0.20
-9	0.22	0.00
-8	0.00	0.20
-6	0.45	0.20
-5	0.00	0.20
-4	0.45	0.59
-3	0.00	0.20
-2	1.12	0.98
-1	1.56	2.75
0	27.23	18.63
1	18.30	13.14
2	24.33	25.29
3	8.48	10.98
4	5.80	9.61
5	3.35	3.92
6	3.79	7.25
7	2.23	2.55
8	0.89	1.37
9	0.22	0.78
10	0.45	0.00
11	0.67	0.20
12	0.45	0.98
Total <i>N</i>	100 (448)	100 (510)

Source: Clingingsmith, Khwaja, and Kremer 2009.

able outcomes, so our method for computing confidence intervals is expected to be reasonably accurate. In order to estimate our 95% interval, we must form a complete schedule of potential outcomes. We add 0.47 to the observed $Y_i(0)$ outcomes in the control group in order to approximate the control group's unobserved $Y_i(1)$ values; we subtract 0.47 from the treatment group's observed $Y_i(1)$ outcomes in order to

approximate the treatment group's unobserved $Y_i(0)$ values. Simulating 100,000 random allocations using this schedule of potential outcomes and sorting the estimated ATEs in ascending order, we find that the 2,500th estimate is 0.16 and the 97,501st estimate is 0.79, so the 95% interval is [0.16, 0.79].

The statistical interpretation of this interval is as follows: over hypothetical replications of this experiment, intervals created in this manner have a 95% chance of bracketing the true ATE. So without other information about the true ATE, we conclude that there is a 95% probability that the interval from 0.16 to 0.79 includes the true ATE.

Substantively, we infer that winning the visa lottery led to an increase in positive feelings toward people from foreign countries. Unfortunately, given the data at hand, there is no easy way to translate the estimated ATE of 0.47 into other metrics that have more tangible meaning in terms of societal outcomes or individual behavior. For example, we do not know how positive responses on this scale translate into cooperative diplomatic relations, increased international trade, or friendly behavior toward visitors from these countries. And because the outcome measure is specific to this study, we are unable to compare the 0.47 effect of this intervention to the effect of other interventions. This gap in our understanding sets the stage for further research. Now that this visa lottery has revealed a causal effect, the next steps are to conduct further visa lottery studies using other outcome measures and to measure the effects of other types of interventions using this survey metric.

When presented with experimental results that are not scaled in relation to interpretable outcome metrics or the effects of other interventions, researchers often fall back on the calculation of *standardized effect size*. This approach compares the estimated effect to the naturally occurring degree of variation in outcomes by dividing the estimated ATE by the standard deviation in the control group.¹⁴ Using this formula, the apparent ATE in this study moves people by about one-fifth of a standard deviation. Again, we confront a problem of interpretation. Is a 0.2 movement in standard deviation big or small? Researchers sometimes invoke rules of thumb: effects of less than 0.3 are considered small, between 0.3 and 0.8 are considered medium, and above 0.8 are considered large.¹⁵ One should be cautious about applying these standards for three reasons. First, the standard deviation is a sample-specific statistic; if one's experimental subjects happen to be Pakistanis who share a similar view of foreigners prior to the intervention, the standard deviation will be small, and the standardized effect will seem large. Second, the standard deviation tends to increase when outcomes are measured with error, as is often the case with survey measures of attitudes. Third, even small standardized effects can be substantively important if they alter a hard-to-move dependent variable. The standard deviation of men's height

is about 2.8 inches, but a dietary supplement that causes a half-inch increase in height would be heralded as remarkable. In much the same way, an intervention that produces a change in attitudes toward foreigners is noteworthy given the difficulty of changing attitudes in this domain.

3.6 Sampling Distributions for Experiments That Use Block or Cluster Random Assignment

The concepts and estimation techniques presented in previous sections may be adapted to experiments in which subjects are assigned randomly but in ways that depart from simple or complete random assignment. In this section, we discuss two such classes of experimental designs: block randomization and cluster randomization.

3.6.1 Block Random Assignment

Block random assignment refers to a procedure whereby subjects are partitioned into subgroups (called blocks or strata), and complete random assignment occurs within each block. For example, suppose we have 20 subjects in our experiment, 10 men and 10 women. Suppose our experimental design calls for 10 subjects to be placed into the treatment condition. If we were to use complete random assignment, chances are that we would end up with unequal numbers of men and women in the treatment group. Block randomization, on the other hand, ensures equal numbers of men and women will be assigned to each experimental condition. First, we partition the subject pool into men and women. From the pool of male subjects, we randomly assign five into the treatment group; from the pool of female subjects, we randomly assign five into the treatment group. In effect, block randomization creates a series of miniature experiments, one per block.

Block randomized designs are used to address a variety of practical and statistical concerns. Sometimes program requirements dictate how many subjects of each type to place in the treatment group. Imagine, for example, that a summer reading program aimed at elementary school students seeks to evaluate its impact on school performance and retention during the following academic year. The school is able to admit only a small fraction of those who apply, and school administrators worry that if too many children with low levels of preparedness are admitted to the program, teachers will find it difficult to manage their classes effectively. These administrators insist that 60% of the children admitted to the program pass an initial test of basic skills. The way to address this concern is by blocking on initial test scores and allocating students within each block so that 60% of the students who are randomly admitted to the program have passed the basic skills test. For example, suppose the school

¹⁴ This standardized statistic is known as Glass's Δ ; from Glass 1976.

¹⁵ Cohen 1988.

BOX 3.8

Number of Possible Assignments under Complete Random Assignment and Blocked Random Assignment

Let $0 < m < N$. Of N observations, m are placed into the treatment group, and $N - m$ are placed into the control group. The number of possible randomizations under complete random assignment is

$$\frac{N!}{m!(N-m)!}$$

For example, the number of randomizations under complete random assignment when $N = 20$ and $m = 10$ is

$$\frac{20!}{10!10!} = 184,756.$$

In order to calculate the number of possible random assignments under a blocked design with B blocks, calculate the number of random allocations in each block: $r_1, r_2, \dots, r_B$. The total number of random assignments is $r_1 \times r_2 \times \dots \times r_B$. For example, when we randomly assign half of the 10 men to treatment and half of the 10 women to treatment, there are

$$\left(\frac{10!}{5!5!}\right)\left(\frac{10!}{5!5!}\right) = 63,504$$

possible random allocations.

can admit 50 of the 100 applicants. Forty of the applicants failed the initial test, and 50 passed. The researcher could create two blocks: one block of students who passed the basic skills test and another block of students who failed. Each block is sorted in random order, and the researcher selects the first 20 students in the block containing those who failed the basic skills test and the first 30 students in the block containing students who passed the test. This procedure ensures that the 60% requirement is satisfied. This design approach comes in handy when resource constraints prevent researchers from treating more than a certain number of subjects from certain regions or when concerns about fairness dictate that treatments be apportioned equally across demographic groups.

Block randomization also addresses two important statistical concerns. First, blocking helps reduce sampling variability. Sometimes the researcher is able to partition the subjects into blocks such that the subjects in each block have similar potential outcomes. For example, students who fail the basic skills test presumably share the same potential outcomes; the same goes for students who pass the basic skills test.

By randomizing within each block, the researcher eliminates the possibility of rogue randomizations that, by chance, place all of the students who fail the basic skills test into the treatment group. Under simple or complete random assignment, these outlandish assignments rarely occur; under block random assignment, they are ruled out entirely. Second, blocking ensures that certain subgroups are available for separate analysis. When analyzing a study involving 10 men and 10 women, a researcher might be interested in comparing the ATE among men to the ATE among women. But what if complete random assignment puts 9 of the 10 men into the treatment group and 9 of the 10 women in the control group? In that case, the treatment effects among men and among women would both be estimated very imprecisely. Blocked randomization guarantees that a specified proportion of a subgroup will be assigned to treatment.

In order to illustrate how blocking works, let's consider a stylized example inspired by Olken's study of corruption in Indonesia.¹⁶ The subjects in this experiment are public works projects, and the treatment is heightened financial oversight by government officials. Outcomes are measured in terms of the amount of money that is unaccounted for (and presumably stolen) when the books are closed on the project. For purposes of illustration, we present in Table 3.3 the complete schedule of potential outcomes for 14 projects, 8 of which are in Region A and 6 in Region B. Because of resource constraints, each region has the capacity to audit only two of its projects. In our example, the ATE is -3 in Region A and -5 in Region B. For both regions combined, the ATE is $(-3)(8/14) + (-5)(6/14) = -3.9$. In general, the relationship between the overall ATE and the ATE within each block j is:

$$ATE = \sum_{j=1}^J \frac{N_j}{N} ATE_j, \quad (3.10)$$

where J is the number of blocks, the blocks are indexed by j , and the weight N_j/N denotes the share of all subjects who belong to block j .

Before studying the statistical precision of the blocked design, it is useful to consider as a point of comparison the precision of a design that uses complete random assignment. Suppose treatments had been assigned to any 4 of the 14 projects through complete random assignment. Equation (3.4) indicates that the true standard error would have been:

$$SE(ATE) = \sqrt{\frac{1}{14-1} \left\{ \frac{(4)(40.41)}{10} + \frac{(10)(32.49)}{4} + (2)(31.03) \right\}} = 3.50. \quad (3.11)$$

In order to calculate the standard error from a blocked design, we must first calculate the standard error within each block. In our stylized example, the projects within each region have similar potential outcomes. As the variances of $Y_i(0)$ and $Y_i(1)$ are

¹⁶ Olken 2007.

TABLE 3.3

Schedule of potential outcomes for public works projects when audited (Y(1)) and not audited (Y(0))

Village	Block	All subjects		Block A subjects		Block B subjects	
		Y(0)	Y(1)	Y(0)	Y(1)	Y(0)	Y(1)
1	A	0	0	0	0		
2	A	1	0	1	0		
3	A	2	1	2	1		
4	A	4	2	4	2		
5	A	4	0	4	0		
6	A	6	0	6	0		
7	A	6	2	6	2		
8	A	9	3	9	3		
9	B	14	12			14	12
10	B	15	9			15	9
11	B	16	8			16	8
12	B	16	15			16	15
13	B	17	5			17	5
14	B	18	17			18	17
Mean		9.14	5.29	4.00	1.00	16.0	11.0
Variance		40.41	32.49	7.75	1.25	1.67	17.0
Cov(Y(0), Y(1))		31.03		2.13		1.00	

much lower within each region than they are when the two regions are combined, the standard error drops markedly when we analyze each region separately. For Region A, the standard error is 1.23; for Region B, it is 2.71. The remaining task is to use these block-specific standard errors to assess the uncertainty of the estimated ATE for subjects in both regions combined. The formula turns out to be straightforward and easily extends to any number of blocks.⁷ For two blocks, the standard error is:

$$SE(\widehat{ATE}) = \sqrt{(SE_1)^2 \left(\frac{N_1}{N}\right)^2 + (SE_2)^2 \left(\frac{N_2}{N}\right)^2}, \quad (3.12)$$

⁷ The formula for any number of blocks is

$$SE(\widehat{ATE}) = \sqrt{\sum_1^J \left(\frac{N_j}{N}\right)^2 SE^2(ATE_j)}.$$

[This formula follows from the general rule about the variance of a sum of independent random variables: $Var(aA + bB) = a^2 Var(A) + b^2 Var(B)$.]

where SE_j refers to the standard error of the estimated ATE in block j , and N_j refers to the number of observations in block j . Filling in the numbers from our example gives a standard error of:

$$SE(\widehat{ATE}) = \sqrt{(1.23)^2 \left(\frac{8}{14}\right)^2 + (2.71)^2 \left(\frac{6}{14}\right)^2} = 1.36. \quad (3.13)$$

The example illustrates the potential benefits of blocking. By making a small design change, we greatly improve the precision with which we estimate the ATE. The standard error plummets from 3.50 to 1.36. The stark difference in sampling distributions is illustrated in Figure 3.1. Under complete random assignment, 141/1001 = 14.1% of the estimated ATEs are greater than zero, which means there is a 14.1% chance that the experiment will indicate that audits are ineffective or actually exacerbate pilfering even though (as we know from Table 3.3) the treatment is effective. Under blocked assignment, just 1/420 = 0.2% of the estimated ATEs are greater than zero.

Let's now consider how one would go about analyzing a block randomized experiment, such as the one reported in Table 3.4, which shows the observed outcomes from a single experiment based on the schedule of potential outcomes from Table 3.3. Estimating the overall ATE is straightforward: first estimate the ATE within

FIGURE 3.1

Sampling distribution under complete randomization (above);
sampling distribution under blocked randomization (below)

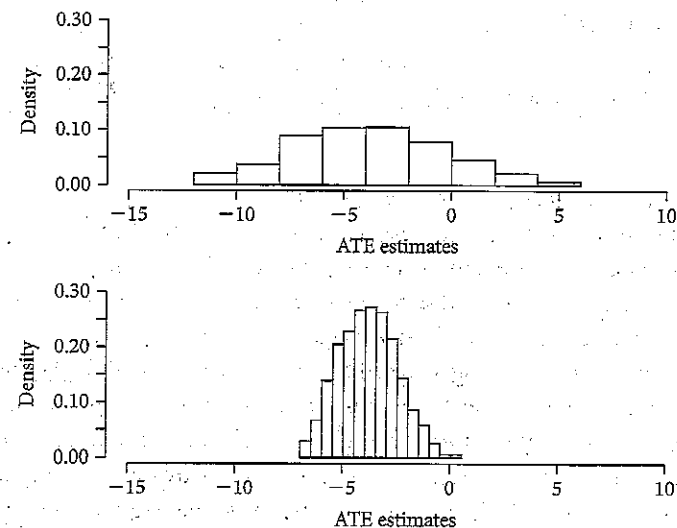


TABLE 3.4

Observed outcomes in a block randomized experiment

All subjects		
Block	Y(0)	Y(1)
A	0	?
A	1	?
A	?	1
A	4	?
A	4	?
A	6	?
A	6	?
A	?	3
B	14	?
B	?	9
B	16	?
B	16	?
B	17	?
B	?	17

each block (denoted $\widehat{ATE}_j$ for each block j) and then compute a weighted average based on the ratio of N_j to N :

$$\begin{aligned}\widehat{ATE} &= (\widehat{ATE}_1) \left(\frac{N_1}{N} \right) + (\widehat{ATE}_2) \left(\frac{N_2}{N} \right) = (-1.5) \left(\frac{8}{14} \right) + (-2.75) \left(\frac{6}{14} \right) \\ &= -2.04.\end{aligned}\quad (3.14)$$

Notice that the weights (N_1/N) and (N_2/N) sum to 1.0.

It should be stressed that this estimator is *not* equivalent in general to a comparison of average outcomes among all subjects in treatment and control. When the probability of being assigned to the treatment group varies by block, as it does in this example, comparing means for all subjects generates a biased estimate of the ATE. In Table 3.4, the probability of being assigned to the treatment group is higher in Region B than in Region A, and Region B tends to have higher average potential outcomes than Region A. Even if there were no treatment effect, if we pool together the two regions, potential outcomes will tend to be higher in the treatment group than in the control group. Unless the probability of assignment to the treatment group is identical for every block, pooling observations across blocks will produce biased estimates

In order to estimate the standard error of the $\widehat{ATE}$, estimate the standard error of each $\widehat{ATE}_j$ using equation (3.6), and insert the block-level estimates of the standard errors into equation (3.12) above:

$$\widehat{SE}(\widehat{ATE}) = \sqrt{(1.43)^2 \left(\frac{8}{14} \right)^2 + (4.05)^2 \left(\frac{6}{14} \right)^2} = 1.92. \quad (3.15)$$

Hypothesis tests and confidence intervals extend directly to blocked designs. When simulating the sampling distribution, remember that randomization takes place within each block. We begin by testing the sharp null hypothesis of no treatment effect. In this example, there are just 420 possible random assignments, 46 of which are less than or equal to the observed estimate of -2.04 , implying a one-tailed p -value of 0.11. Confidence intervals may be approximated by simulating the full schedule of potential outcomes within each block by adding the $\widehat{ATE}_j$ to the control group's outcomes and subtracting the $\widehat{ATE}_j$ from the treatment group's outcomes. The 95% interval when treatment effects are assumed to be constant extends from -4.82 to 0.62 . One might wonder about the accuracy of this method of estimating the 95% confidence intervals, given that we know from the potential outcomes in Table 3.3 that the constant effects assumption is false. It turns out that applying this estimation procedure to each of the 420 possible random assignments produces an estimated confidence interval that brackets the true ATE in 94.8% of the random allocations. In this example, the estimated 95% confidence intervals do in fact cover the ATE with probability of approximately 0.95.

3.6.1.1 MATCHED PAIR DESIGN

Notice that in order to calculate $\widehat{SE}(\widehat{ATE}_j)$ for each block, every block must contain at least two observations in treatment and two observations in control. This requirement will not be met by *matched pair designs*, experiments in which every block contains just two subjects, one of which is assigned to treatment. Fortunately, analysis of matched pair experiments is straightforward. The ATE is estimated by subtracting the control outcome from the treatment outcome in each block and averaging over all blocks. (Because the probability of assignment to the treatment group is 0.5 for all blocks, one obtains identical estimates by subtracting the average outcome among the $N/2$ subjects in the control condition from the average outcome among the $N/2$ subjects in the treatment condition). These block-level differences are then used to estimate the standard error:

$$\widehat{SE}(\widehat{ATE}) = \sqrt{\frac{1}{N \left(\frac{N}{2} - 1 \right)} \sum_{j=1}^J (\widehat{ATE}_j - \widehat{ATE})^2}$$

The analysis of matched pairs is readily integrated into experiments that contain blocks of varying size. For example, suppose that, as presented in Table 3.5, assignment in the public works audit experiment had taken place within two matched pairs (Blocks B and C) and one larger block containing 10 subjects (Block A). In each matched pair, one subject is assigned to treatment.

The matched pairs generate an $\widehat{ATE}_j$ of -3 and 0, respectively. The larger block generates an $\widehat{ATE}_j$ of -2.6. The estimate of the overall ATE is:

$$\widehat{ATE} = \left(\frac{2}{14}\right)(-3) + \left(\frac{2}{14}\right)(0) + \left(\frac{10}{14}\right)(-2.6) = -2.3. \quad (3.17)$$

In order to calculate the standard error, we first calculate the standard error for the matched pair part of the experiment using equation (3.16):

$$\widehat{SE}(\widehat{ATE}) = \sqrt{\frac{1}{\frac{4}{2}\left(\frac{4}{2}-1\right)}}((-3 - (-1.5))^2 + (0 - (-1.5))^2) = 1.5, \quad (3.18)$$

TABLE 3.5

Observed outcomes in a block randomized experiment with blocks of varying size

All subjects		
Block	Y[0]	Y[1]
A	0	?
A	1	?
A	?	1
A	4	?
A	4	?
A	6	?
B	6	?
B	?	3
A	14	?
A	?	9
A	16	?
A	16	?
C	17	?
C	?	17

and then combine the standard error for the matched pair part of the design with the standard errors from the other block using equation (3.12):

$$\widehat{SE}(\widehat{ATE}) = \sqrt{(1.5)^2 \left(\frac{4}{14}\right)^2 + (4.64)^2 \left(\frac{10}{14}\right)^2} = 3.3. \quad (3.19)$$

3.6.1.2 SUMMARY OF THE ADVANTAGES AND DISADVANTAGES OF BLOCKING

Blocking provides a way to address two classes of design objectives. The first arises when practical or ethical imperatives dictate the number of treatments that may be assigned to certain groups of subjects. Blocking allows the researcher to build a randomized experiment around these constraints. The second set of objectives has to do with statistical precision. When blocks are formed by grouping subjects with similar potential outcomes, substantial improvements in precision are possible, particularly in small samples. Chapter 4 discusses how one might form blocks based on background information available at the time of randomization.

Does blocking have a downside? Although it is possible to construct perverse examples in which blocking harms precision, blocking rarely has negative consequences in practice.¹⁸ Even blocking subjects into random subsets (which does nothing to group subjects according to their potential outcomes) is an experimental design that is no worse, in terms of the precision with which the ATE is calculated, than complete random assignment. In practice, the biggest downside associated with blocking is the risk that the data will be misanalyzed. The ATE must be estimated block by block when the probability of treatment assignment varies by block, and care must be taken to conduct hypothesis tests in a manner that follows the exact procedure by which the randomization was carried out.

Sometimes block randomization is not feasible. It is not uncommon for field experiments to be conducted under severe time constraints, and background information necessary to form blocks may be unavailable at the time of random allocation. Similarly, the implementation of randomization within blocks may exceed the technical capacity of those implementing random allocation in field settings. Failure to block randomize should not be regarded as a serious design flaw, especially in studies where the treatment and control groups contain more than 100 subjects.¹⁹ As we will see in the next chapter, if you do not use blocking, you can do almost as well in terms of precision through statistical adjustments.

¹⁸ Imai 2008.

¹⁹ Rosenberger and Lachin 2002.

3.6.2 Cluster Random Assignment

To this point, we have considered only experiments in which each of the subjects is *individually* assigned to the treatment or control group. To visualize the difference between individual- and cluster-level assignment, imagine an experiment on the persuasive influence of TV commercials. Suppose a credit card company has purchased records for 200,000 people in each of five cities. A researcher proposes to randomly assign commercials to two of the five cities and assess purchasing behavior among the 1,000,000 people in the database. The N of individuals is 1,000,000, but there are only five clusters. If simple random assignment were used to allocate individuals to treatment or control, the sampling distribution would be very narrow, because chance differences between experimental groups would tend to balance out due to the very large number of subjects in each group. But under cluster assignment, all subjects in the same cluster are placed as a group into either the treatment or control condition; in effect, cluster assignment rules out all of the possible allocations in which individuals in the same cluster are assigned to different experimental conditions. Because all of the inhabitants of a city are assigned as a cluster to treatment or control, the attributes they share are placed into one of the experimental groups. If these cities' potential outcomes differ appreciably, random assignment of just five clusters will produce unbiased but imprecise estimates.

For practical reasons, cluster assignment is sometimes unavoidable. In the case of TV broadcasts, it is currently impossible to administer the treatment to specific individuals; instead, one purchases advertising that targets geographic clusters. The same goes for treatments that are administered by canvassers or election monitors, who may be able to travel only to a few general locations.

Sampling distributions may change markedly when clusters, rather than individuals, are assigned to experimental groups. For the moment, let's consider the simplest scenario: each of the clusters contains the same number of subjects. When clusters are the same size, the usual difference-in-means estimator remains unbiased. Cluster assignment, however, will generate more sampling variability than complete random assignment if subjects within the same cluster tend to share similar potential outcomes.

In order to illustrate the consequences of within-cluster similarity, Table 3.6 presents a schedule of potential outcomes for 12 classrooms located in schools that have varying levels of academic achievement. The average treatment effect in this example is 4 points on a standardized test. Each school contains three classrooms. Using a cluster design, the experimenter randomly assigns two of the four schools to the treatment group. The total number of possible random assignments is:

$$\frac{4!}{2!2!} = 6. \tag{3.20}$$

Differences-in-means based on cluster assignment yield unbiased estimates of the treatment effect: the average estimate is 4. To obtain the standard error, we calculate

TABLE 3.6
Hypothetical schedule of potential outcomes for 12 classrooms in four schools—high sampling variability

School	Classroom	Classroom-level potential outcomes		Cluster-level mean potential outcomes	
		$Y(0)$	$Y(1)$	$\bar{Y}(0)$	$\bar{Y}(1)$
A	A-1	0	4	1	5
A	A-2	1	5		
A	A-3	2	6		
B	B-1	2	6	3	7
B	B-2	3	7		
B	B-3	4	8		
C	C-1	3	7	4	8
C	C-2	4	8		
C	C-3	5	9		
D	D-1	7	11	8	12
D	D-2	8	12		
D	D-3	9	13		

the standard deviation across all possible random assignments and find it to be 2.9. Suppose, however, that our experiment had instead allocated the 12 classrooms using complete random assignment, half to treatment and half to control. The standard error would have been substantially smaller. The number of possible randomizations is

$$\frac{12!}{6!6!} = 924, \tag{3.21}$$

and across them the standard error is 1.6. Clustering, in other words, led to an 81% increase in the standard error, which is roughly the same increase that would have occurred if our sample size had declined from 12 to 4.

Why were the consequences of clustering so severe? The schools in the preceding example were quite different in terms of their classrooms' average potential outcomes; meanwhile, the classrooms within each school were relatively similar. Let's see what happens when we rearrange the example so that schools are more similar in terms of average classroom performance. Table 3.7 shows another example with four schools that have three classrooms apiece. The classrooms are the same as in Table 3.6, but they are situated in different schools. Across the six possible randomizations of a cluster assignment, the average estimated treatment effect is 4 with a standard error of 0.57, which is actually lower than under complete random assignment.

TABLE 3.7
Hypothetical schedule of potential outcomes for 12 classrooms in four schools—low sampling variability

School	Classroom	Classroom-level potential outcomes		Cluster-level mean potential outcomes	
		$Y_j(0)$	$Y_j(1)$	$\bar{Y}_j(0)$	$\bar{Y}_j(1)$
A	A-1	0	4	4	8
A	A-2	3	7		
A	A-3	9	13		
B	B-1	2	6	4	8
B	B-2	3	7		
B	B-3	7	11		
C	C-1	1	5	3.3	7.3
C	C-2	4	8		
C	C-3	5	9		
D	D-1	4	8	4.7	8.7
D	D-2	8	12		
D	D-3	2	6		

The bottom line is that the penalty associated with clustering depends on the variability of the cluster-level means. This point is driven home by the formula for the true standard error assuming a clustered design with equally sized clusters. Equation (3.22) expresses this standard error in a manner that parallels equation (3.4):

$$SE(\widehat{ATE}) = \sqrt{\frac{1}{k-1} \left\{ \frac{m \text{Var}(\bar{Y}_j(0))}{N-m} + \frac{(N-m) \text{Var}(\bar{Y}_j(1))}{m} + 2 \text{Cov}(\bar{Y}_j(0), \bar{Y}_j(1)) \right\}}, \quad (3.22)$$

where $\bar{Y}_j(0)$ is the average untreated potential outcome in the j th cluster, $\bar{Y}_j(1)$ is the average treated potential outcome in the j th cluster, and k is the number of clusters. If one could somehow create clusters that had nearly identical average potential outcomes, $\text{Var}(\bar{Y}_j(0))$ and $\text{Var}(\bar{Y}_j(1))$ would be small, and the ATE would be estimated with great precision. Unfortunately, researchers are usually stuck with highly variable cluster means and therefore large standard errors. Cluster means tend to vary because clusters tend to be formed based on geography, institutions, age groups, and so forth, and subjects within the same cluster tend to have similar potential outcomes. Comparing equations (3.4) and (3.22) reveals why cluster randomized designs often have lower power than completely randomized designs. Unlike equation (3.4), which divides the elements inside the braces by $N - 1$, equation (3.22) divides by $k - 1$.

In terms of data analysis, the techniques for testing hypotheses and forming confidence intervals must be adjusted when analyzing clustered experiments. Adapting hypothesis testing to clustered experiments is straightforward. Under the sharp null hypothesis, treated and untreated potential values are identical, so the researcher observes the full schedule of potential outcomes. In order to test the sharp null hypothesis, simulate all possible assignments of clusters to treatment. The sampling distribution of estimated ATEs under the sharp null hypothesis will usually be much wider under clustered assignment than under complete random assignment. Confidence intervals may be obtained by creating a complete schedule of potential outcomes under the assumption of constant treatment effects. Again, simulate all possible random assignments of clusters in order to obtain the sampling distribution.²⁰ To obtain standard errors that tend to err on the conservative side, use the formula

$$SE(\widehat{ATE}) = \sqrt{\frac{N \text{Var}(\bar{Y}_j(0))}{k(N-m)} + \frac{N \text{Var}(\bar{Y}_j(1))}{km}}, \quad (3.23)$$

where k is the number of clusters. When attempting to improve the precision of a cluster randomized design, researchers should look for ways to increase the number of clusters; increasing the number of subjects per cluster typically has little effect on the standard error because it does little to reduce the variance of the cluster-level potential outcomes.

To this point, we have considered the relatively simple case in which clusters contain the same number of subjects. Analysis becomes more complicated when clusters differ in size. If cluster size covaries with potential outcomes, the usual difference-in-means estimator is biased.²¹ This bias disappears as the number of clusters increases. Because the full set of potential outcomes is unobserved, the threat of bias is something that can only be assessed indirectly. A straightforward diagnostic approach is to examine whether the estimated ATE appears to change when we restrict attention to large or small clusters.²² If one suspects bias, an alternative estimator focuses on the difference in total outcomes:

$$\widehat{ATE} = \frac{k_c + k_T}{N} \left(\frac{\sum Y_i(1) | d_i=1}{k_T} - \frac{\sum Y_i(0) | d_i=0}{k_c} \right), \quad (3.24)$$

²⁰ Because estimation of the ATE consumes a degree of freedom, one applies a correction when forming the upper (and lower) bound of the confidence interval. The width of the interval expands by $\sqrt{(k-1)/(k-2)}$, where k is the number of clusters.

²¹ Middleton and Aronow 2011. The intuition behind this bias is that random assignment of unequal size clusters allows m to vary from one random allocation to the next, which creates bias when cluster size is related to potential outcomes. See exercise 12.

²² Recall that the difference-in-means estimator is unbiased when cluster size is constant, so partitioning the data by cluster size produces approximately unbiased estimates. The drawback to this approach, however, is that subsets of a dataset may lack power to detect a relationship between cluster size and potential outcomes.

where k_c is the number of clusters assigned to control and k_t is the number of clusters assigned to treatment. Unlike the difference-in-means estimator, the estimator equation (3.24) divides by quantities (k_t , k_c , and N) that are unaffected by which units are assigned to treatment. The estimator in equation (3.24) is equivalent to difference-in-means when clusters are all the same size and is unbiased for any number of clusters or arrangement of cluster sizes; its primary drawback is that it produces estimates with more sampling variability than the conventional difference-in-means estimator because it ignores the number of individual subjects in each cluster.²³ If there are a few clusters with large numbers of subjects and distinctive potential outcomes, estimates may vary markedly depending on the experimental condition to which they are assigned.

In order to illustrate these statistical techniques, let's examine a cluster randomized voter mobilization experiment conducted in Kansas City prior to its 2003 municipal elections.²⁴ A group called ACORN targeted 28 low-income precincts in an effort to mobilize voters on behalf of a ballot measure designed to fund municipal bus service. ACORN wanted to work within selected precincts in order to make it easier to train and supervise its canvassers. Of the 28 precincts in the sample, 14 were randomly allocated to the treatment group, and ACORN made repeated attempts to canvass and call voters on its target list in those precincts. The 28 precincts contain a total of 9,712 voters, and the number of targeted voters per precinct ranges from 31 to 655; the marked difference in cluster size leaves open the possibility for bias if precincts with large numbers of potential ACORN sympathizers have different potential outcomes from precincts with relatively few.

We begin by estimating the effect using a difference-in-means approach. Turnout was 33.5% in the treatment group ($N = 4,933$) and 29.1% in the control group ($N = 4,779$), implying an ATE of 4.4 percentage points. In order to test the sharp null hypothesis, we generate a random subset of 100,000 of the 40,116,600 possible random assignments of the 28 precincts and find that 5.0% of the randomizations yield estimates of the ATE that are greater than or equal to 4.4. Thus, the estimated one-tailed p -value is 0.05. We estimate a 95% confidence interval around the estimate of 4.4 by using the constant treatment effect assumption to fill in the schedule of potential outcomes: $(-0.5, 9.3)$. At the low end of this interval, the campaign failed to increase turnout; at the high end, it had a profound effect on turnout. This is the price one pays for a clustered design. The interval is approximately 2.7 times wider than it would have been had researchers randomly assigned individual voters rather than precincts. Fortunately, as we will see in the next chapter, this confidence interval narrows considerably when we control for the voter turnout rates of these subjects in previous elections.

²³ See Middleton and Aronow 2011. The standard error for the difference-in-totals estimator tends to be large because the variance of a total (as opposed to an average) tends to be large.
²⁴ Arceneaux 2005.

Next, we repeat the analysis using a difference-in-totals approach. Applying equation (3.24) to these data yields an estimated ATE of:

$$\frac{14+14}{9712} \left(\frac{1654}{14} - \frac{1392}{14} \right) = 0.054, \quad (3.25)$$

or 5.4 percentage points, which is slightly larger than the estimate obtained using a difference-in-means. This estimate, however, is subject to greater sampling variability. Over a random subset of 100,000 of the 40,116,600 possible random assignments under the sharp null hypothesis, 20% of the estimates are greater than 5.4 percentage points, implying a p -value of 0.20. Assuming a constant treatment effect of 5.4 percentage points, we obtain a 95% confidence interval that ranges from -6 to 17 . The loss of precision associated with the difference-in-totals estimator conveys a useful design lesson: when conducting a cluster randomized experiment, try to work with clusters of similar size. If that is not possible, block on cluster size so that the conventional difference-in-means estimator may be used within blocks without risk of bias.

SUMMARY

Sampling variability is a constant concern for the experimental researcher. During the design phase, decisions about how to select subjects, allocate them to experimental conditions, and measure outcomes may have important repercussions for the precision of the experimental estimates. This chapter has considered some of the basic principles of experimental design, and later chapters will build on this framework as we grapple with the complications that arise due to noncompliance and attrition.

The standard error is a measure of statistical uncertainty. A close examination of standard error formulas provides useful design insights. Among design ideas discussed in this chapter are the potential advantages of assigning similar numbers of subjects to treatment and control conditions and of eliminating extraneous sources of variability in outcomes. Comparing the standard errors of different unbiased estimators is another source of insight. Blocked random assignment can improve precision when the blocking variables strongly predict outcomes; cluster assignment may diminish precision when units with similar potential outcomes wind up in the same cluster.

Once the data are collected, sampling variability represents one of the foremost concerns in the interpretation of results. Randomized experiments, in principle, generate empirical results that have two interpretations: either the treatment exerted a causal effect, or the result occurred due to sampling variability. Randomization inference is a valuable method for assessing the sampling variability of estimates. Under the sharp null hypothesis, the observed data reveal all of the potential outcomes,

providing the researcher with access to the full sampling distribution. By simulating results from all possible randomizations, the researcher can examine the sampling distribution of any test statistic. Of special interest is the sampling distribution of the estimated average treatment effect and other test statistics that might indicate heterogeneous treatment effects. The latter topic is discussed in detail in Chapter 9.

The use of simulation to conduct hypothesis tests is a flexible approach that may be tailored to fit the details of any random allocation. For example, researchers may divide subjects into blocks and randomly allocate subjects within each block to treatment or control. Or researchers may combine subjects into clusters and allocate all of the members of a cluster to treatment or control. These subtleties of design sometimes profoundly affect the resulting sampling distribution, but the same basic hypothesis testing tools apply. Under the sharp null hypothesis of no treatment effect for any subject, the full schedule of potential outcomes is observed, allowing the researcher to simulate the sampling distribution by following the exact procedures used to assign subjects to experimental conditions.

The estimation of confidence intervals is more challenging because it relies on assumptions about each subject's treatment effect. The method described here invokes the assumption that treatment effects are constant, and that the ATE is equal to the estimated ATE. This assumption works well for experimental designs in which the same number of subjects are in treatment and control conditions. In most cases, the estimated 95% confidence intervals will be conservative in the sense that they have more than a 95% chance of bracketing the true ATE. When applied to designs where one experimental group is much larger than the other, the estimated confidence intervals may be too wide or too narrow depending on whether outcomes in the two groups have different variances. In ambiguous cases, researchers may wish to compare their estimated confidence intervals to the intervals generated by other estimation procedures.

SUGGESTED READINGS

Freedman, Pisani, and Purves (2007) offers a good discussion of the differences between observational and experimental research; see its appendix for a derivation of equation (3.4). See Imai (2008) on the sampling distribution of matched pair experiments. Murray (1998) and Boruch (2005) discuss cluster randomized experiments.

EXERCISES: CHAPTER 3

See <http://isps.research.yale.edu/FEDAI> for datasets and programs for conducting randomization inference.

1. Important concepts:

- What is a standard error? What is the difference between a standard error and a standard deviation?

- How is randomization inference used to test the sharp null hypothesis of no effect for any subject?
 - What is a 95% confidence interval?
 - How does complete random assignment differ from block random assignment and cluster random assignment?
 - Experiments that assign the same number of subjects to the treatment group and control group are said to have a "balanced design." What are some desirable statistical properties of balanced designs?
- Rewrite equation (3.4) substituting for $Y_i(1)$ using the equation $Y_i(1) = Y_i(0) + \tau_i$. Assume that $N = 2m$, and interpret the implications of the resulting formula for experimental design.
 - Using the equation $Y_i(1) = Y_i(0) + \tau_i$, show that when we assume that treatment effects are the same for all subjects, $\text{Var}(Y_i(0)) = \text{Var}(Y_i(1))$ and the correlation between $Y_i(0)$ and $Y_i(1)$ is 1.0.
 - Consider the schedule of outcomes in the table below. If treatment A is administered, the potential outcome is $Y_i(A)$, and if treatment B is administered, the potential outcome is $Y_i(B)$. If no treatment is administered, the potential outcome is $Y_i(0)$. The treatment effects are defined as $Y_i(A) - Y_i(0)$ or $Y_i(B) - Y_i(0)$.

Subject	$Y_i(0)$	$Y_i(A)$	$Y_i(B)$
Miriam	1	2	3
Benjamin	2	3	3
Helen	3	4	3
Eva	4	5	3
Billie	5	6	3

Suppose a researcher plans to assign two observations to the control group and the remaining three observations to just one of the two treatment conditions. The researcher is unsure which treatment to use.

- Applying equation (3.4), determine which treatment, A or B, will generate a sampling distribution with a smaller standard error.
 - What does the result in part (a) imply about the feasibility of studying interventions that attempt to close an existing "achievement gap"?
- Using Table 2.1, imagine that your experiment allocates one village to treatment.
 - Calculate the estimated difference-in-means for all seven possible randomizations.
 - Show that the average of these estimates is the true ATE.
 - Show that the standard deviation of the seven estimates is identical to the standard error implied by equation (3.4).
 - Referring to equation (3.4), explain why this experimental design has more sampling variability than the design in which two villages out of seven are assigned to treatment.

- (e) Explain why, in this example, a design in which one of seven observations is assigned to treatment has less sampling variability than a design in which six villages out of seven are assigned to treatment.

The Clingingsmith, Khwaja, and Kremer study discussed in section 3.5 may be used to test the sharp null hypothesis that winning the visa lottery for the pilgrimage to Mecca had no effect on the views of Pakistani Muslims toward people from other countries. Assume that the Pakistani authorities assigned visas using complete random assignment. Conduct 10,000 simulated random assignments under the sharp null hypothesis. How many of the simulated random assignments generate an estimated ATE that is at least as large as the actual estimate of the ATE? What is the implied one-tailed p -value? How many of the simulated random assignments generate an estimated ATE that is at least as large in *absolute value* as the actual estimate of the ATE? What is the implied two-tailed p -value?

A diet and exercise program advertises that it causes everyone who is currently starting a diet to lose at least seven pounds more than they otherwise would have during the first two weeks. Use randomization inference (the procedure described in section 3.4) to test the hypothesis that $\tau_i = 7$ for all i . The treatment group's weight losses after two weeks are {2, 11, 14, 0, 3} and the control group's weight losses are {1, 0, 0, 4, 3}. In order to test the hypothesis $\tau_i = 7$ for all i using the randomization inference methods discussed in this chapter, subtract 7 from each outcome in the treatment group so that the exercise turns into the more familiar test of the sharp null hypothesis that $\tau_i = 0$ for all i . When describing your results, remember to state the null hypothesis clearly, and explain why you chose to use a one-sided or two-sided test.

Naturally occurring experiments sometimes involve what is, in effect, block random assignment. For example, Titiunik studies the effect of lotteries that determine whether state senators in Texas and Arkansas serve two-year or four-year terms in the aftermath of decennial redistricting.⁵ These lotteries are conducted within each state, and so there are effectively two distinct experiments on the effects of term length. An interesting outcome variable is the number of bills (legislative proposals) that each senator introduces during a legislative session. The table below lists the number of bills introduced by senators in both states during 2003.

- For each state, estimate the effect of having a two-year term on the number of bills introduced.
- For each state, estimate the standard error of the estimated ATE.
- Use equation (3.10) to estimate the overall ATE for both states combined.
- Explain why, in this study, simply pooling the data for the two states and comparing the average number of bills introduced by two-year senators to the average number of bills introduced by four-year senators leads to biased estimates of the overall ATE.
- Insert the estimated standard errors into equation (3.12) to estimate the standard error for the overall ATE.
- Use randomization inference to test the sharp null hypothesis that the treatment effect is zero for senators in both states.

Texas		Arkansas	
Term Length: 0 = four-year term; 1 = two-year term	# of bills introduced	Term Length: 0 = four-year term; 1 = two-year term	# of bills introduced
0	18	0	11
0	29	0	15
0	41	0	23
0	53	0	24
0	60	0	25
0	67	0	26
0	75	0	28
0	79	0	31
0	79	0	33
0	88	0	34
0	93	0	35
0	101	0	35
0	103	0	36
0	106	0	38
0	107	0	52
0	131	0	59
1	29	1	9
1	37	1	10
1	42	1	14
1	45	1	15
1	45	1	15
1	54	1	17
1	54	1	18
1	58	1	19
1	61	1	19
1	64	1	20
1	69	1	21
1	73	1	23
1	75	1	23
1	92	1	24
1	104	1	28
		1	30
		1	32
		1	34

Camerer reports the results of an experiment in which he tests whether large, early bets placed at horse tracks affect the betting behavior of other bettors.²⁶ Selecting pairs of long-shot horses running in the same race whose betting odds were approximately the same when betting opened, he placed two \$500 bets on one of the two horses approximately 15 minutes before the start of the race. Because odds are determined based on the proportion of total bets placed on each horse, this intervention causes the betting odds for the treatment horse to decline and the betting odds of the control horse to rise. Because Camerer's bets were placed early, when the total betting pool was small, his bets caused marked changes in the odds presented to other bettors. (A few minutes before each race started, Camerer cancelled his bets.) While the experimental bets were still "live," were other bettors attracted to the treatment horse (because other bettors seemed to believe in the horse) or repelled by it (because the diminished odds meant a lower return for each wager)? Seventeen pairs of horses in this study are listed below. The outcome measure is the number of dollars that were placed on each horse (not counting Camerer's own wagers on the treatment horses) during the test period, which begins 16 minutes before each race (roughly 2 minutes before Camerer began placing his bets) and ends 5 minutes before each race (roughly 2 minutes before Camerer withdrew his bets).

- One interesting feature of this study is that each pair of horses ran in the same race. Does this design feature violate the non-interference assumption, or can potential outcomes be defined so that the non-interference assumption is satisfied?
- A researcher interested in conducting a randomization check might assess whether, as expected, treatment and control horses attract similarly sized bets prior to the experimental intervention. Use randomization inference to test the sharp null hypothesis that the bets had no effect prior to being placed.
- Calculate the average increase in bets during the experimental period for treatment horses and control horses. Compare treatment and control means, and interpret the estimated ATE.
- Show that the estimated ATE is the same when you subtract the control group outcome from the treatment group outcome for each pair, and calculate the average difference for the 17 pairs.
- Use randomization inference to test the sharp null hypothesis of no treatment effect for any subject. When setting up the test, remember to construct the simulation to account for the fact that random assignment takes place within each pair. Interpret the results of your hypothesis test and explain why a two-tailed test is appropriate in this application.

Suppose that 800 students were randomly assigned to classrooms of 25 students apiece, and these classrooms were then randomly assigned as clusters to treatment and control. Assume that non-interference holds. Use equations (3.4) and (3.22) to explain why this clustered design has the same standard error as complete random assignment of individual students to treatment and control.

	Total bets on treatment horse 16 minutes before race	Total bets on treatment horse 5 minutes before race (excluding experimental bets)	Treatment change	Total bets on control horse 16 minutes before race	Total bets on control horse 5 minutes before race	Control change	Difference in changes
Pair 1	\$533	\$1,503	\$970	\$587	\$2,617	\$2,030	-\$1,060
Pair 2	\$376	\$1,186	\$810	\$345	\$1,106	\$761	\$49
Pair 3	\$576	\$1,366	\$790	\$653	\$2,413	\$1,760	-\$970
Pair 4	\$1,135	\$1,666	\$531	\$1,296	\$2,260	\$964	-\$433
Pair 5	\$158	\$367	\$209	\$201	\$574	\$373	-\$164
Pair 6	\$282	\$542	\$260	\$269	\$489	\$220	\$40
Pair 7	\$909	\$1,597	\$688	\$775	\$1,825	\$1,050	-\$362
Pair 8	\$566	\$933	\$367	\$629	\$1,178	\$549	-\$182
Pair 9	\$0	\$555	\$555	\$0	\$355	\$355	\$200
Pair 10	\$330	\$786	\$456	\$233	\$842	\$609	-\$153
Pair 11	\$74	\$959	\$885	\$130	\$256	\$126	\$759
Pair 12	\$138	\$319	\$181	\$179	\$356	\$177	\$4
Pair 13	\$347	\$812	\$465	\$382	\$604	\$222	\$243
Pair 14	\$169	\$329	\$160	\$165	\$355	\$190	-\$30
Pair 15	\$41	\$297	\$256	\$33	\$75	\$42	\$214
Pair 16	\$37	\$71	\$34	\$33	\$121	\$88	-\$54
Pair 17	\$261	\$485	\$224	\$282	\$480	\$198	\$26

²⁶ Camerer 1998. This example draws on the second of Camerer's studies and restricts the sample to cases which a treatment horse is compared to a single control horse.

- Use the data in Table 3.3 to simulate cluster randomized assignment.
- Suppose that clusters are formed by grouping observations {1,2}, {3,4}, {5,6}, ..., {13,14}. Use equation (3.22) to calculate the standard error assuming half of the clusters are randomly assigned to the treatment.
 - Suppose that clusters are instead formed by grouping observations {1,14}, {2,13}, {3,12}, ..., {7,8}. Use equation (3.22) to calculate the standard error assuming half of the clusters are randomly assigned to the treatment.
 - Why do the two methods of forming clusters lead to different standard errors? What are the implications for the design of cluster randomized experiments?
2. Below is a schedule of potential outcomes for six classrooms, which are located in three schools. Using a cluster randomized design, researchers will assign one of the three schools (and all the classrooms it contains) to the treatment group.
- What is the average treatment effect among the six classrooms?
 - There are three possible randomizations. Is the difference-in-means estimator unbiased?
 - In general, cluster random assignment generates biased results when (i) clusters vary in size, (ii) potential outcomes vary by cluster, and (iii) the number of clusters is too small to ensure that m of N units are placed into the treatment condition in each randomization. Show what happens in this example when School A and School B are combined for purposes of random assignment, so that there is a 0.5 probability that either School C is placed in treatment or Schools A and B are placed in treatment. Does this design yield unbiased estimates? What are the implications of this exercise for the design of cluster randomized experiments?

School	Classroom	$Y_i[0]$	$Y_i[1]$
A	A-1	0	0
B	B-1	0	1
B	B-2	0	1
C	C-1	0	2
C	C-2	0	2
C	C-3	0	2

APPENDIX

A3.1 Power

Before running an experiment, the researcher may be interested in calculating the experiment's *power*. In the context of experimental research, power typically refers to the probability that the researcher will be able to reject the null hypothesis of no treatment effect. However, power analysis requires some guesswork: the researcher must furnish values of unknown parameters, such as the size of the true ATE.

To illustrate a power analysis, consider a completely randomized experiment where $N/2$ of N units are selected into a binary treatment. The researcher must now make assumptions about the distributions of outcomes for treatment and for control units. In this example, the researcher assumes that the control group has a normally distributed outcome with mean μ_c , the treatment group has a normally distributed outcome with mean μ_t , and both groups' outcomes have a standard deviation σ . The researcher must also choose α , the desired level of statistical significance (typically 0.05).

Under this scenario, there exists a simple asymptotic approximation for the power of the experiment (assuming that the significance test is two-tailed):

$$\beta = \Phi\left(\frac{|\mu_t - \mu_c| \sqrt{N}}{2\sigma} - \Phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right), \quad (\text{A3.1})$$

where β is the statistical power of the experiment, $\Phi(\cdot)$ is the normal cumulative distribution function (CDF), and $\Phi^{-1}(\cdot)$ is the inverse of the normal CDF. These functions are easily implemented using R or even a spreadsheet; Microsoft Excel's syntax for $\Phi(\cdot)$ is `normsdist()`, and its syntax for $\Phi^{-1}(\cdot)$ is `normsinv()`. For example, if $N = 500$, $\mu_c = 60$, $\mu_t = 65$, $\sigma = 20$ and $\alpha = 0.05$, the power of the experiment, β , would be approximately 0.80. This calculation implies that the proposed experiment has an 80% probability of generating results that lead to the rejection of the null in the presence of a true treatment effect.

Inspection of the formula reveals that power rises as sample size N goes up. One answer to a lack of power is to expand the sample. Power also increases with effect size $|\mu_t - \mu_c|$; strengthening the treatment is another way to remedy insufficient power. Power also increases as σ diminishes, a fact that encourages researchers to reduce noise, obtain prognostic covariates (see Chapter 4), and minimize heterogeneity among subjects.

When assessing experimental designs for which analytical power analysis is not feasible, the researcher may use computer simulation to estimate power. The researcher posits a schedule of potential outcomes for all experimental units. Then, M simulated experiments are generated where treatment is allocated to these units, and p -values are generated for each simulated experiment. The estimated power of the experiment is then the proportion of all M simulated treatment allocations such that